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Project Report

The ReSDDP Model

Principles and Algorithm

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ABSTRACT
This report documents the principles behind the open source long-term hydropower scheduling model ReSDDP. The model uses the stochastic dual dynamic programming algorithm in combination with additional decomposition techniques to facilitate the computation of long-term strategies for use of hydropower in the context of a renewable power system.

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Chapter 1

Introduction

This report documents the principles behind the open source long-term hydropower scheduling model ReSDDP. It was prepared as a part of the research project "Modeling a 100% Renewable Electricity System" (RES100) [1]. The computer code is maintained at the GitLab repository `https://gitlab.sintef.no/energy/res100/resddp`. The model reads data prepared for the EMPS model (version 10).

The main goal of the ReSDDP model is to find the expected marginal value of water (*water value*) in large-scale hydropower systems as part of renewable electricity systems. Supplementary functionality for simulating the dispatch by using the obtained water values is available, but will not be discussed in detail here. The model is designed to explicitly model costs and technical constraints related to system dispatch. Traditionally, this type of model is classified as long-term hydrothermal scheduling models [1], emphasizing the balance between operation of long-term reservoirs and use of thermal generation in lieu of inflow uncertainty. In future scenarios with large-scale deployment of variable renewable energy sources (VRES), such as wind and solar power, and a gradual phase-out of dispatchable fossil-fueled generation technologies, the key contributors to this balance change. In addition to VRES, new technologies are expected to add flexibility on both the supply and demand side. These changes will stress the need to update long-term scheduling models with proper representations of the short-term variability and uncertainty related to VRES as well as technical descriptions of the new technologies.

The ReSDDP model is based on stochastic optimization, where both long- and short-term uncertainties can be represented. It was designed to balance detailed system description and computational effort by use of decomposition techniques, as described below and illustrated in Fig. 1.1. Decomposition techniques enable us to formulate a lightweight and computationally efficient function to estimate coupled water values for hydropower subsystems. The decomposition allows the separation of:

- **A Core Function:** The core function consists of a stochastic dual dynamic programming (SDDP) algorithm applied to hydropower subsystems. The SDDP model expresses hydropower strategies by means of a *future expected cost function (FCF)*, from which water values can be derived. Both the detailed hydropower system and the residual demand for hydropower are represented by use of decomposition techniques.
- **Approximate Hydropower:** A detailed formulation of the hydropower system is formulated so that the feasibility of aggregated decisions per subsystem can be controlled. The feasible operation region of each hydropower subsystem is constrained by *feasibility spaces*, as elaborated in Chapter 3 and indicated in Fig. 1.1. The feasibility space is expressed as a function of a subsystem's decision variables, involving capabilities to store and produce energy, ramp production and provide reserve capacity.
- **Residual Demand for Hydropower:** A detailed description of the least-cost dispatch problem is used to derive the residual demand cost function seen by the hydropower system, as elaborated in Chapter 4

[1] https://www.sintef.no/en/projects/2023/modeling-a-100-renewable-electricity-system-res100/

and indicated in Fig. 1.1. The residual demand cost function is expressed as a function of hydropower subsystem decision variables.

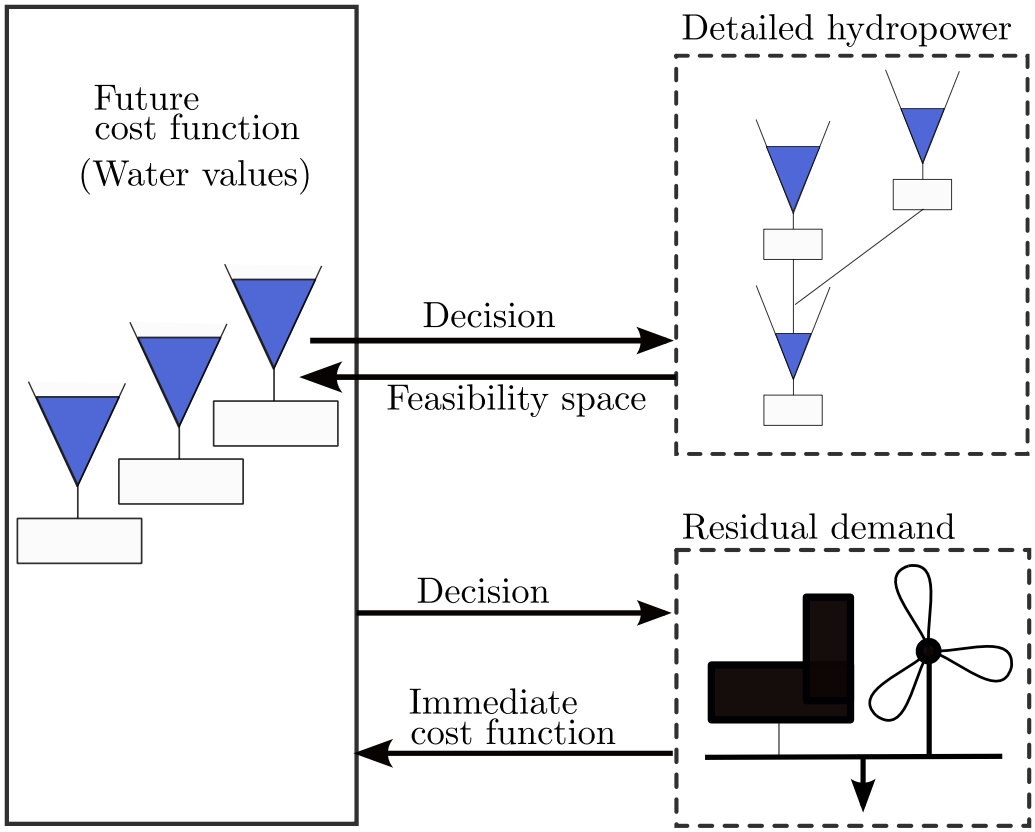


Figure 1.1: Illustration of the spatial decomposition process.

The report is structured as follows. First, the problem formulation and the SDDP solution algorithm is described in Chapter 2. The approximation of the capabilities of a hydropower subsystem by feasibility spaces is elaborated in Chapter 3. Then an interface between the hydropower subsystems and the remaining power system (the "residual demand for hydropower") is elaborated on in 4. A summary of the ReSDDP model principles and the underlying assumptions is provided in Chapter 5. Finally, ReSDDP model was tested on a small-scale test dataset and the results, in terms of computational performance and water values, are presented and discussed in Chapter 6.

Chapter 2

Problem Formulation

As a starting point we describe the scheduling problem and how the SDDP algorithm can be applied to solve it. All details regarding the hydropower system, other technologies on the supply and demand side, and the transmission grid are explicitly formulated in a *least-cost dispatch problem*. Applying the complete formulation to a large-scale system will quickly lead to a computationally intractable problem. This motivates the need for decomposition techniques that are further described in this report.

The SDDP algorithm was introduced in [2] and has later been widely applied in operative scheduling models and has been subject to improvements and extensions by the research community. An introduction to some of the extensions and application of the algorithm are provided in Section 2.6.

2.1 The Scheduling Problem

A planning horizon of multiple years is assumed, consisting of $t = 1, \dots, T$ weekly *decision stages*, where each stage covers a decision period for which we can assume that the uncertainty is known. In the following we assume one stage to represent one week. A finer time discretization of the week, consisting of $k = 1, \dots, K$ *time steps*, allows decisions to adapt to variability in net demand.

The long-term hydropower scheduling problem is defined in (2.1), where x_t are the state variables $y_t = [y_{t1}, \dots, y_{tK}]$ the stage variables for each decision stage t and time step k. The stage variables represent the operational decisions to be made in each stage and time step, while *state variables* transfer information about the system state between stages. One seeks an operating strategy minimizing the expected cost of supplying electricity in (2.1a), while accounting for the end-of-horizon valuation of stored water in $\Psi(x_T)$, and respecting all relevant technical, environmental, and juridical constraints in (2.1b)-(2.1d).

$$\min_{(x_1,y_1),\dots,(x_T,y_T)} \mathbb{E}\left\{\sum_{t=1}^{T} f_t(x_t, y_t) + \Psi(x_T)\right\} \tag{2.1a}$$

$$\text{s.t.} \quad \mathbf{W}x_t + \mathbf{H}x_{t-1} + \mathbf{G}y_t = h(\xi_t) \tag{2.1b}$$

$$\mathbf{B}y_t = 0 \tag{2.1c}$$

$$(x_t, y_t) \in X_t \tag{2.1d}$$

$$\forall t \in \{1, 2, \dots, T\}$$

The constraints are indicated in (2.1b)-(2.1d), where the initial state vector x_0 is given, X_t is the feasible set for the decision variables of decision stage t, and $\mathbf{W}$, $\mathbf{H}$, $\mathbf{G}$, and $\mathbf{B}$, are matrices of suitable dimensions. The expectation in (2.1a) is taken over the stochastic variables. Examples of uncertainties that could be considered over the scheduling horizon are inflow, wind and solar power, demand, and exogenous prices.

The problem in (2.1) can be classified as a multi-stage stochastic optimization problem, which may be efficiently solved by decomposition techniques [3]. We assume that the mathematical model can be expressed in terms of linear (or at least convex) functional relationships, so that the optimization problem(s) can be solved by efficient linear programming (LP) algorithms. Moreover, we assume that the stochastic problem has slack variables to ensure feasible solutions when decomposing it (relatively complete recourse).

2.2 Solution Strategy

This work applies the SDDP algorithm, which is a sampling-based variant of multi-stage Benders decomposition. The problem in (2.1) can be decomposed into stage-wise nested LP problems of type (2.2). A detailed description of the stage problem is provided later in (2.7).

$$\mathcal{Q}_t(x_{t-1}) = \min_{x_t, y_t} f_t(x_t, y_t) + \alpha_{t+1}(x_t) \tag{2.2a}$$

$$\text{s.t.} \quad (x_t, y_t) \in X_t(x_{t-1}, \xi_t) \tag{2.2b}$$

The SDDP algorithm is iterative and typically relies on statistical convergence criteria. Each main SDDP iteration consists of a *forward simulation* and a *backward recursion*, repeatedly solving stage-wise decision problems of type (2.2). The variable α_{t+1} represent the future expected cost function (FCF) seen from the beginning of stage $t+1$ (or similarly, the end of stage t). This variable will gradually be constrained by Benders cuts constructed in the SDDP backward iterations. We will refer to these Benders cuts as *FCF cuts* in this report. The SDDP forward iterations provide simulated state space trajectories corresponding to a set of sampled inflow scenarios. We assume uncertainty in inflow to the hydropower reservoirs, and that this inflow can be described by a vector autoregressive model of first order (VAR1).

2.2.1 Forward Simulation

In the SDDP forward simulation a sequence of inflow scenarios $s = 1, \dots, NS$ covering the period of analysis from $t = 1, \dots, T$ are found by sampling from the VAR1 inflow model. Subsequently, the stage decision problem detailed in (2.7) is solved for each stage t along the simulated scenario s, results are collected, and state variables are updated for the next stage. The simulated state at the end of the stage is used as the initial state for the next stage. The forward simulation provides an updated set of state trajectories. Moreover, the forward simulation is used to obtain a lower bound J^- in (2.3) representing the first stage cost f_1 plus the future expected cost α_2 seen from the beginning of the second stage, and an upper bound J^+ in (2.4) representing the average simulated cost, assuming equiprobable scenarios.

$$J^- = f_1(x_1, y_1) + \alpha_2 \tag{2.3}$$

$$J^+ = \frac{1}{NS} \sum_{s=1}^{NS} \sum_{t=1}^{T} f_t(x_{st}, y_{st}) \tag{2.4}$$

Uncertainty around the upper bound can be used as a convergence check. One computes the standard deviation sd as in (2.5) and checks if the lower bound (J^-) lies in the interval $\left[J^+ - 1.96\frac{sd}{\sqrt{NS}}; J^+ + 1.96\frac{sd}{\sqrt{NS}}\right]$.

$$sd^2 = \frac{1}{NS-1} \sum_{s=1}^{NS} \left(\sum_{t=1}^{T} f_t(x_{st}, y_{st}) - J^+ \right)^2 \tag{2.5}$$

For practical purposes the stabilization of the lower bound may serve as an alternative criterion.

2.2.2 Backward Recursion

Benders cuts at the end of the planning horizon $t = T$ can be obtained from a predefined final value function Ψ. For $t = T-1, \dots, 1$ in the backward recursion one loops through each state trajectory obtained from the last forward iteration. Starting from the state at the end of stage $t-1$, for each realization of stochastic variables one computes the optimal operation for stage t by solving (2.7). From the sensitivities of the objective function to the initial state values, found as dual values from (2.7b) and (2.7c), new FCF cuts of type (2.7j) at the end of stage $t-1$ are obtained.

2.3 Stochastic Model

The stochastic model for energy inflow is briefly described in the following. As a starting point, inflow data are read for the detailed hydropower system. Inflow to the reservoir (regulated) and the power station (unregulated) are specified for each hydropower module. Unique historical inflow records may be associated with both the regulated and unregulated inflow. These records are scaled according to an expected annual inflow to form time series of regulated and unregulated inflow for each hydropower module.

A simple aggregation, based on the best efficiency to sea, is used to aggregate detailed inflow per module, measured in water, to time series of *energy inflow* per subsystem. In this process, regulated and unregulated inflows are accumulated to energy inflow that enters the subsystem energy reservoir. The unregulated part defines a minimum generation requirement for the subsystems equivalent power station. Next, energy inflow time series are normalized [1], and a VAR1 model is fitted to the normalized data. The VAR1 model takes the form

$$\mathbf{z}_t = \mathbf{\Phi} \mathbf{z}_{t-1} + \epsilon_t \quad , \tag{2.6}$$

where $\mathbf{z_t}$ and $\mathbf{z_{t-1}}$ are vectors of normalized energy inflows to each subsystem for time stages t and $t-1$, respectively, $\mathbf{\Phi}$ is the correlation matrix, and ϵ_t is the vector of residuals ("white noise").

2.4 Detailed Stage Problem

In this section we provide a mathematical description of the stage problem behind the high-level description in (2.2). This detailed stage problem is formulated as an LP problem and is solved for time stage t both in the forward simulation and backward recursion in the standard SDDP formulation. As will be discussed in depth in Chapters 3 and 4, the ReSDDP model further decomposes the stage problem to separate the treatment of detailed hydropower and the residual market.

To ease formulation, a combination of vectors and matrices (in bold) and their corresponding elements are used. Units are in $m3/s$, Mm^3, or k€. Conversion factors Γ_w and Γ_e ensure consistent treatment of units. We use the sets $\mathcal{H}$ for hydropower modules (reservoir+plant); $\mathcal{K}$ for time steps; $\mathcal{A}$ for price areas (or 'bidding zones'); $\mathcal{G}_a$ for generators in area a; $\mathcal{D}_a$ for demands in area a; $\omega_h^{D,B,S}$ for upstream hydro modules discharging (D), bypassing (B), and spilling (S) to hydro module h; $\mathcal{N}_h$ for efficiency-curve segments for hydro module h; and $\mathcal{L}_a^{+/-}$ for transmission lines directed out of (+) or in to (−) area a. For simplicity we assume existence of a VAR1 inflow model for the detailed system in (2.7), using the same notation as for the aggregated inflow model in Section 2.3.

$$\mathcal{Q}_t(\mathbf{v}^*_{t-1}, \mathbf{z}^*_{t-1}) =$$
$$\min \sum_{k\in\mathcal{K}} \sum_{a\in\mathcal{A}} \left[\sum_{g\in\mathcal{G}_a} C^G_{gt} p^G_{gk} - \sum_{d\in\mathcal{D}_a} C^D_{dt} p^D_{dk} + C^R p^R_{ak} \right] + \alpha_{t+1} \tag{2.7a}$$

[1] A normalized value is obtained by subtracting the mean and dividing by the standard deviation.

$$\mathbf{v}_{t-1} = \mathbf{v}^*_{t-1} \quad : \boldsymbol{\pi}^v \tag{2.7b}$$

$$\mathbf{z}_{t-1} = \mathbf{z}^*_{t-1} \quad : \boldsymbol{\pi}^z \tag{2.7c}$$

$$\mathbf{z}_t = \boldsymbol{\Phi}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t \tag{2.7d}$$

$$I_{ht} = \sigma_{ht} z_{ht} + \mu_{ht} \quad \forall h \tag{2.7e}$$

$$\begin{aligned} &v_{hk} - v_{h,k-1} + \Gamma_w\left(q^D_{hk} + q^B_{hk} + q^S_{hk}\right) \\ &- \Gamma_w\Big(\sum_{j\in\omega^D_h} q^D_{jk} + \sum_{j\in\omega^B_h} q^B_{jk} + \sum_{j\in\omega^S_h} q^S_{jk}\Big) = \zeta_k I_{ht} \end{aligned} \quad \forall h,k \tag{2.7f}$$

$$q^D_{hk} = \sum_{n\in\mathcal{N}_h} q^D_{nhk} \quad \forall h,k \tag{2.7g}$$

$$p^H_{hk} = \Gamma_e \sum_{n\in\mathcal{N}_h} \eta_{nh} q^D_{nhk} \quad \forall h,k \tag{2.7h}$$

$$\begin{aligned} &\sum_{h\in\mathcal{H}_a} p^H_{hk} + \sum_{g\in\mathcal{G}_a} p^G_{gk} - \sum_{d\in\mathcal{D}_a} p^D_{dk} + p^R_{at} \\ &\quad + \sum_{\ell\in\mathcal{L}^+_a} f_{\ell k} - \sum_{\ell\in\mathcal{L}^-_a} f_{\ell k} \geq D_{ak} - W_{ak} \end{aligned} \quad \forall a,k \tag{2.7i}$$

$$\alpha_{t+1} + \sum_{h\in\mathcal{H}} \pi^v_{hct} v_{ht} + \sum_{h\in\mathcal{H}} \pi^z_{hct} z_{ht} \geq \beta_{ct} \quad \forall c \tag{2.7j}$$

$$0 \leq q^D_{nhk} \leq \overline{Q}^D_{nht} \quad \forall n,h \tag{2.7k}$$

$$0 \leq q^B_{hk} \leq \overline{Q}^B_{ht} \quad \forall h \tag{2.7l}$$

$$\underline{V}_{ht} \leq v_{hk} \leq \overline{V}_{ht} \quad \forall h \tag{2.7m}$$

$$0 \leq p^G_{gk} \leq \overline{P}^G_{gk} \quad \forall g,k \tag{2.7n}$$

$$0 \leq p^D_{dk} \leq \overline{P}^D_{dk} \quad \forall d,k \tag{2.7o}$$

$$0 \leq p^R_{ak} \leq \overline{D}_{ak} \quad \forall a,k \tag{2.7p}$$

$$-\overline{F}_{\ell t} \leq f_{\ell tk} \leq \overline{F}_{\ell t} \quad \forall \ell,k \tag{2.7q}$$

The objective in (2.7a) seeks to minimize cost of operating the system within the decision period, including power generation p^G at marginal cost C^G, price-sensitive consumption p^D at marginal value C^D, curtailment of price-insensitive demand p^R at marginal cost C^R, an the future expected cost of operation expressed by the variable α_{t+1}.

The state variables are the vector of initial reservoir volumes $\mathbf{v}^*_{\mathbf{t-1}}$ and inflow state z^*_{t-1}, so that $\mathbf{x}_{t-1} = [\mathbf{v}^*_{\mathbf{t-1}}, \mathbf{z}^*_{\mathbf{t-1}}]$. Constraints (2.7b)-(2.7c) present variables v_{t-1} and z_{t-1} as copies of the corresponding observed state variables. These constraints are included to ease the computation of dual values $\boldsymbol{\pi}^v$ and $\boldsymbol{\pi}^z$.

The vector of normalized inflows z_t for all detailed reservoirs and for the current decision period is found in (2.7d) according to the AR1 inflow model, where $\boldsymbol{\epsilon}_t$ represents a sample. The physical inflow is obtained in (2.7e). Reservoir balances for each hydropower reservoir h and time step k are presented in (2.7f). Reservoir volumes v are balanced with inflow and decisions related to inflow and outflow expressed by three separate waterways: Discharge q^D, bypass q^B and spillage q^S. Hydropower production is represented per module by a piecewise linear conversion between water and energy in (2.7h), where the discharge decision is segmented in (2.7g), with efficiencies η.

Power balances for each price area a and time step k are provided in (2.7i), balancing production from hydropower p^H and thermal p^G generation with price-sensitive p^D and price-insensitive D demand, wind power W, curtailment p^R, and the exchange with connected price areas f_ℓ.

The linear constraints (FCF cuts) shown in (2.7j) are generated in the backward recursion of the SDDP al-

gorithm and express the future expected cost α_{t+1} as a function state variables v_{ht} and z_{ht}. Finally, variable bounds are shown in (2.7k)-(2.7q).

2.5 Need for Approximation

The problem described in (2.7) provides a basic mathematical formulation of the stage problem which is solved repeatedly in the SDDP forward simulation and backward recursion. More functionality is typically included, such as environmental constraints related to hydropower operation, power flow constraints, constraints limiting the ramping capability of various production technologies, other types flexibility such as batteries and demand response, etc. Including all details soon lead to stage problems that can be classified as medium- to large-scale LP problems, comprising hundreds of thousands of variables and constraints. Relevant experiences with detailed formulation of the unit commitment and dispatch problem for the Nordic power system are presented in [4–6], emphasizing the complexity in this type of problem.

In a typical SDDP run, the stage problem detailed in (2.7) is solved millions of times in total. One can view the stage problem as a function that is visited frequently to provide: (i) reservoir trajectories during the forward simulation; and (ii) estimate the marginal value of the state variables in the backward recursion. In the subsequent chapters we will describe how one can define a lightweight function by making use of decomposition principles. Approximations leading to this lightweight function are derived from the underlying detailed optimization problem.

If the main purpose of using SDDP is to estimate coupled water values for the long-term hydropower storages that are later to be used in more detailed simulations of the system, it is reasonable to assume that the detailed formulation in (2.7) can be relaxed or approximated to reduce computational effort. However, such approximations should be carefully assessed in the context of the rapidly evolving power system landscape, e.g, involving more variability and short-term uncertainty. Although the trade-off between modeling accuracy and computational performance is not new to long-term scheduling models, the system changes ahead will challenge us to revisit previous findings in this matter. In this context, there is a need for transparent approximations for which the accuracy can be controlled.

2.6 Further Reading on SDDP

Since the seminal works in [2, 7], the SDDP algorithm has been improved in various directions and applied in numerous studies of hydro-dominated power systems around the world. We provide some pointers to the literature below, while emphasizing that this list by no means is complete.

- Stochastic programming and decomposition techniques are elaborated in [3], e.g., explaining the multi-stage Benders decomposition algorithm in detail. An early reference on multi-stage Benders decomposition including a numerical example on hydrothermal scheduling is provided in [8], while a higher-level description of SDDP is provided in [9].
- Convergence properties are studied by many authors, such as in [10] and [11]. The latter references also discusses limitations regarding the statistical convergence criterion.
- Representation of uncertainty in inflow within SDDP is discussed in [12], emphasizing techniques for avoiding generation of negative inflows within the AR-1 model. Joint treatment of uncertainty in inflow and wind is discussed in [13] and [14].
- The hydropower scheduling model ProdRisk, which is maintained by SINTEF Energy, is widely used for regional (or medium-term) hydropower scheduling in the Nordic market (price-taker assumption). ProdRisk is based on a combination of stochastic dynamic programming and SDDP, treating the power price a stochastic exogenous variable [15, 16].

Chapter 3

Approximating Detailed Hydropower

For a typical description of the Nordic power system, a significant share of variables and constraints in stage problem instances from (2.7) are due to the hydropower system.

We describe a detailed hydropower system as a collection of hydropower *modules* which may be linked to form one or more cascades. A module then forms a basic building block for the hydropower system, including a reservoir and a power station. As an example, consider a model representing a system with 1000 hydropower modules, with hourly time resolution. This would lead to 168.000 reservoir constraints of type (2.7f) and 168.000 hydropower production variables (p^H_{hk}). Moreover, the dimensionality of the FCF represented by cuts of type (2.7j) will be 1000 plus the number of individual inflow time series being represented in the AR1 model. The high dimensionality may hamper both the SDDP convergence properties and the accuracy of the stochastic inflow model, as demonstrated and discussed in [17].

In the following we discuss how the operation of complex hydropower systems can be approximated by use of feasibility spaces, represented by linear inequalities which we refer to as *feasibility cuts*. The details related to the construction and use of feasibility cuts were first described in [18] and later explored in [19]. We use the term hydropower *subsystem* to describe a collection of hydropower modules for which aggregate decision are made. For each price area with hydropower we require at least one hydropower subsystem. The use of feasibility spaces is based on the assumption that operation-dependent costs associated with hydropower operation can be neglected.

3.1 Derivation of Feasibility Cuts

Assume that we represent a hydropower subsystem by its equivalent generation p^H_k and final storage v, both expressed in terms of energy. The state of the subsystem is given by its initial energy volume V_0 and energy inflow I. We would like to know if the subsystem decisions $y = (v, p^H_k)$ are feasible for the given state $z = (V_0, I)$. The answer is given by a function

$$f_p(y,z) = \min s, \qquad Ax \geq b_1(z) \quad : \kappa, \qquad Bx + s \geq b_2(y) \quad : \gamma. \tag{3.1}$$

The constraint set in (3.1) involves decisions x related to the detailed hydropower system, including reservoir balances and production functions detailed in (2.7) and targets for at least meeting subsystem decisions y. The function $f_p(y,z)$ provides a feasibility check: Is it possible for the subsystem in state z to meet requirements provided by the decisions y without using slack variables s? If so, $f_p(y,z) = 0$.

The dual problem of (3.1) can be formulated as:

$$f_d(y,z) = \max\Big(\kappa b_1(z) + \gamma b_2(y)\Big), \qquad \kappa x \leq 0, \qquad \gamma x \leq s. \tag{3.2}$$

Note that the feasible region of (3.2) does not depend on (y,z). Therefore, it is possible to define the feasible region independently of (y,z) by vertices in a finite set $\mathcal{P}$. The optimal solution of (3.2) will always

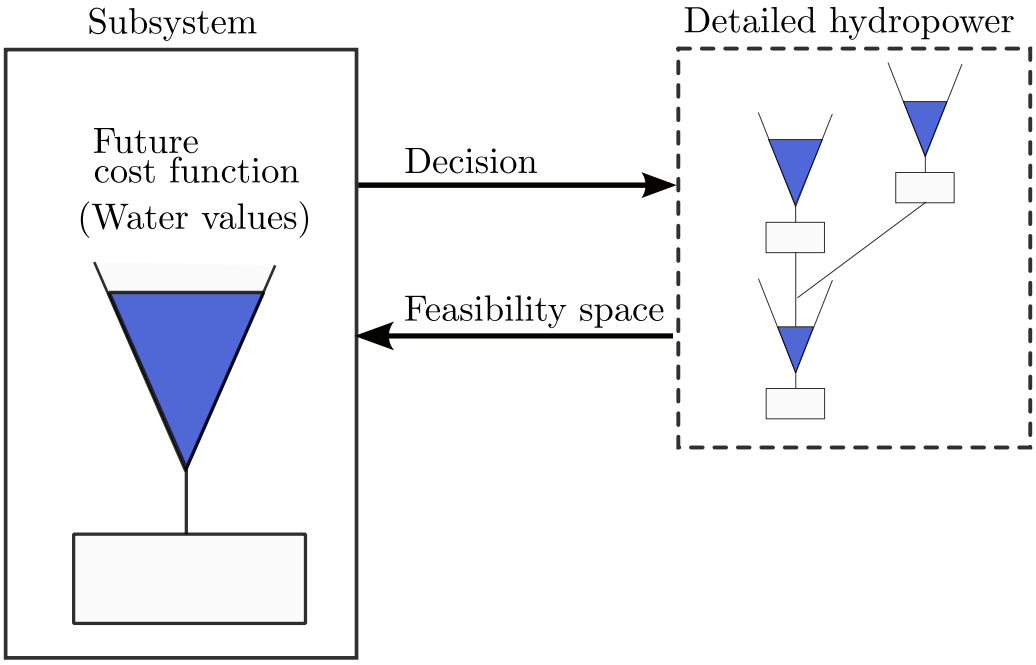


Figure 3.1: Illustration of the spatial decomposition process between aggregate and detailed hydropower

correspond to a vertex of the feasible region, and thus (3.2) can in theory be solved by enumeration

$$f_d(y,z) = \max\Big(\kappa^i b_1(z) + \gamma^i b_2(y)\Big), \qquad \forall i \in \mathcal{P}. \tag{3.3}$$

An equivalent formulation of (3.3) is

$$f(y,z) = \min \alpha, \qquad \alpha \geq \kappa^i b_1(z) + \gamma^i b_2(y), \qquad \forall i \in \mathcal{P}. \tag{3.4}$$

For a convex problem, the optimal solution of (3.4) is equal to (3.3) which is equal to (3.2) and (3.1). In (3.4), α can be considered as the cost of operating the detailed hydropower system. Assuming that this cost is negligible and setting $\alpha = 0$, we can arrive at the set of constraints describing the feasibility space of a subsystem, i.e., possible outcomes of decisions y provided state information z:

$$0 \geq \kappa^i b_1(z) + \gamma^i b_2(y), \qquad \forall i \in \mathcal{P}. \tag{3.5}$$

Inserting for $y = (v, p_k^H)$ and $z = (V_0, I)$ and reorganizing gives a linear inequality (or feasibility cut) on the form:

$$\gamma^v v + \sum_{k \in \mathcal{K}} \gamma_k^p p_k^H + \kappa^v V_0 + \kappa^i I \leq 0. \tag{3.6}$$

All feasibility cuts corresponding to the set $\mathcal{P}$ can in theory be explicitly included in the dispatch problem to constrain aggregated hydropower. Treating such constraints by relaxation is, however, more computationally efficient. That is, we ignore them initially, and add them whenever $f_p^* > 0$. From the feasibility check in (3.1) we record the objective f_p^*, the decisions (v^*, p_k^{H*}), the states (V_0^*, I^*), and the dual values $(\gamma^v, \gamma_k^p, \kappa^v, \kappa^i)$ to form a feasibility cut

$$f^* + \gamma^v (v - v^*) + \sum_{k \in \mathcal{K}} \gamma_k^p (p_k^H - p_k^{H*}) + \kappa^v (V_0 - V_0^*) + \kappa^i (I - I^*) \leq 0. \tag{3.7}$$

The idea is to gradually approximate the feasibility space by adding feasibility cuts of type (3.7) for various realizations of y and z. Eventually feasibility cuts in (3.7) can be simplified to

$$\gamma^v v + \sum_{k \in \mathcal{K}} \gamma_k^p p_k^H + \kappa^v V_0 + \kappa^i I \leq \beta^F \quad , \tag{3.8}$$

where β^F expresses the feasibility cut right-hand side. A collection of constraints of type (3.8) can be seen as a *feasibility space* limiting the decision variables in y provided a state z.

3.1.1 Illustrative Example

To illustrate the basic properties of hydropower aggregation by use of feasibility cuts, we present an example system comprising one hydropower module (reservoir+station) serving a demand in one time period, as illustrated in Fig. 3.2. The demand function is shown in blue in the figure, where unit e represents energy and mu/e represents 'moneteray units' per 'energy'. The hydropower production function displays energy output as a fcuntion of water usage w, where the conversion efficiency is 2.0 e/w for production up to 40 e and 1.0 e/w for production between 40 and 50 e. The maximum reservoir capacity is 100 w and the initial storage is 50 w. We assume that the system has no inflow and that the water value is 110 mu/e.

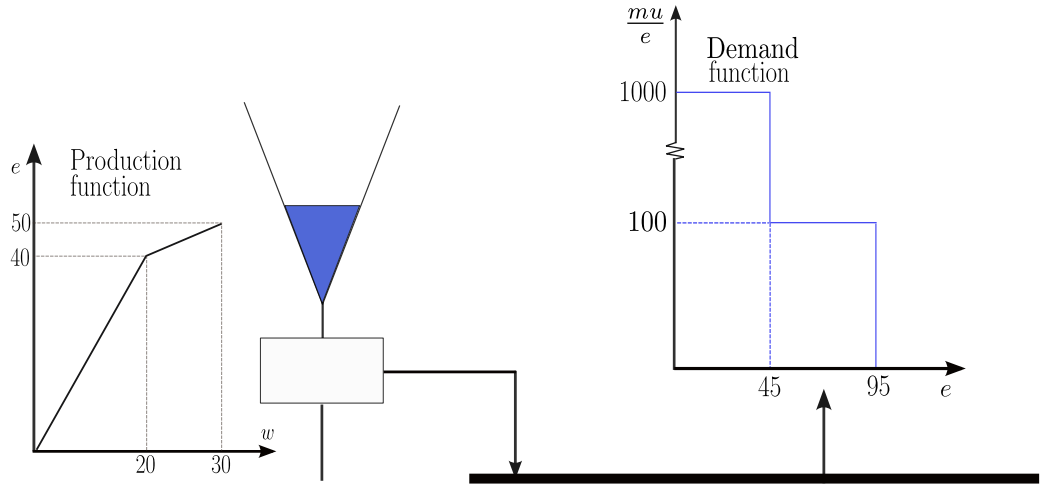


Figure 3.2: Example system.

Consider the hydropower system as a subsystem where properties are expressed in terms of energy, referred to the best efficiency (2.0 e/w). The subsystem's initial storage $v_0 = 100e$, maximum production $\overline{P}^H = 50e$, and maximum storage $\overline{V} = 200e$. The target subsystem production is $45e$ to cover the most valuable demand segment, storing the remaining water for future use.

Fig. 3.3 illustrates the feasibility space for the subsystem decisions, as feasibility cuts c_1 and c_2 are added. First, optimal subsystem decisions ($v^* = 200e, p^{H*} = 45e$) are found. These decisions do not respect the reservoir balance, and thus feasibility cut c_1 is added. The second optimal decision ($v^* = 55e, p^{H*} = 45e$) is then found, as illustrated by the rightmost red circle in Fig. 3.3. Consulting the detailed hydropower description one finds that the production of $45e$ requires discharge at the lowest efficiency segment, and thus using more water than was estimated using the subsystem formulation. In other words, the second optimal decision is not feasible. Thus, feasibility cut c_2 is added, forcing the third optimal decision to ($v^* = 50e, p^{H*} = 45e$), which corresponds to the decision obtained when optimizing the detailed system.

For this simple example there was no need to define a subsystem, but its simplicity allow us to emphasize some important points:

- In the above approach we first overestimate the capabilities of the subsystem and then iteratively constrain the feasibility space of subsystem decisions through consultations with the detailed hydropower representation.
- By initially constraining the subsystem model, the number of feasibility cuts can be reduced. In the case above, cut c_1 can be avoided if the subsytem storage capability is constrained by an energy balance.

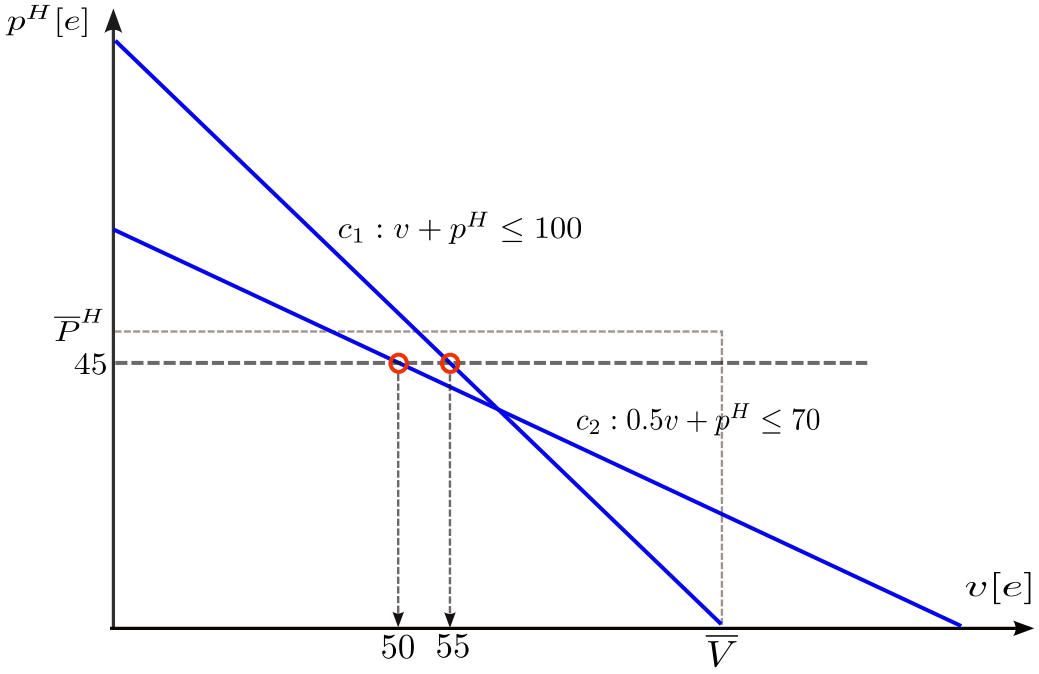


Figure 3.3: Feasibility space visualized.

3.2 Stage Problem with Feasibility Spaces

If we replace the detailed hydropower description per *module* in (2.7) by an aggregated description per *subsystem*, each subsystem's decisions can be constrained by a feasibility space. To limit computational effort, we seek to represent these feasibility spaces in few dimensions. Thus, it will be beneficial to avoid decisions per time step in the feasibility cuts, but rather represent these as aggregated decisions per stage. This paves the way for using different time resolutions when evaluating the subsystems and the corresponding detailed hydropower systems. Rather than hydropower generation per time step p_k^H in (3.8), we can accumulate hydro production per decision stage in (3.9) and reformulate the feasibility cuts in (3.10):

$$e = \sum_{k \in \mathcal{K}} \tau_k p_k^H \tag{3.9a}$$

$$\sum_{k \in \mathcal{K}} \tau_k \underline{P}_k^H \leq e \leq \sum_{k \in \mathcal{K}} \tau_k \overline{P}_k^H \tag{3.9b}$$

$$\gamma^v v + \gamma^e e + \kappa^v V_0 + \kappa^i I \leq \beta \tag{3.10}$$

Detailed hydropower in (2.7) can now be replaced by subsystem decisions limited by feasibility spaces represented by cuts of type (3.10), as expressed in (3.11) below. The decision variables representing end of stage storage (v_{st}) and hydro production per stage (e_{st}) enter feasibility cuts in (3.11i)

$$\mathcal{Q}_t(\mathbf{v}^*_{t-1}, \mathbf{z}^*_{t-1}) =$$

$$\min \sum_{k\in\mathcal{K}} \sum_{a\in\mathcal{A}} \left[\sum_{g\in\mathcal{G}_a} C^G_{gt} p^G_{gk} - \sum_{d\in\mathcal{D}_a} C^D_{dt} p^D_{dk} + C^R p^R_{ak} \right] + \alpha_{t+1} \tag{3.11a}$$

$$\mathbf{v}_{t-1} = \mathbf{v}^*_{t-1} \quad : \boldsymbol{\pi}^v \tag{3.11b}$$

$$\mathbf{z}_{t-1} = \mathbf{z}^*_{t-1} \quad : \boldsymbol{\pi}^z \tag{3.11c}$$

$$\mathbf{z}_t = \boldsymbol{\Phi}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t \tag{3.11d}$$

$$I_{st} = \sigma_{ht} z_{st} + \mu_{st} \quad \forall s \tag{3.11e}$$

$$v_{st} + \sum_{k\in\mathcal{K}} p^H_{sk} + s_{st} = v_{s,t-1} + I_{st} \quad \forall s \tag{3.11f}$$

$$p^H_{sk} + \sum_{g\in\mathcal{G}_a} p^G_{gk} - \sum_{d\in\mathcal{D}_a} p^D_{dk} + p^R_{at} + \sum_{\ell\in\mathcal{L}^+_a} f_{\ell k} - \sum_{\ell\in\mathcal{L}^-_a} f_{\ell k} \geq D_{ak} - W_{ak} \quad \forall a, k \tag{3.11g}$$

$$e_{st} = \sum_{k\in\mathcal{K}} \tau_k p^H_{sk} \quad \forall s \tag{3.11h}$$

$$\gamma^v_{cst} v_{st} + \gamma^e_{cst} e_{st} + \kappa^v_{cst} V_{s,t-1} + \kappa^i_{cst} I_{st} \leq \beta^F_{ct} \quad \forall s, c \tag{3.11i}$$

$$\alpha_{t+1} + \sum_{s\in\mathcal{S}} \pi^v_{sct} v_{st} + \sum_{s\in\mathcal{S}} \pi^z_{sct} z_{st} \geq \beta_{ct} \quad \forall c \tag{3.11j}$$

$$\underline{P}^H_{sk} \leq p^H_{sk} \leq \overline{P}^H_{sk} \quad \forall s, k \tag{3.11k}$$

$$\sum_{k\in\mathcal{K}} \tau_k \underline{P}^H_{sk} \leq e_{st} \leq \sum_{k\in\mathcal{K}} \tau_k \overline{P}^H_{sk} \quad \forall s \tag{3.11l}$$

$$0 \leq p^G_{gk} \leq \overline{P}^G_{gk} \quad \forall g, k \tag{3.11m}$$

$$0 \leq p^D_{dk} \leq \overline{P}^D_{dk} \quad \forall d, k \tag{3.11n}$$

$$0 \leq p^R_{ak} \leq \overline{D}_{ak} \quad \forall a, k \tag{3.11o}$$

$$-\overline{F}_{\ell t} \leq f_{\ell tk} \leq \overline{F}_{\ell t} \quad \forall \ell, k \tag{3.11p}$$

3.3 Feasibility Space Extensions

One can think of the hydropower subsystem as an equivalent with multiple capabilities, such as:

- Energy storage
- Energy production
- Ramping capability
- Reserve capacity procurement

While the two first (energy storage and production) are represented in (3.11), extensions are needed to capture ramping capability and reserve capacity procurement delivered by a hydropower subsystem.

3.3.1 Ramping Capability

When an energy target is passed from the subsystem to the detailed hydropower system for feasibility check, the detailed system will seek to meet the target by minimum use of water. With the formulation presented so far, there are no other subsystem constraints on intra-stage production that limits the detailed system operation. Thus, the subsystem model may suggest profiles $[p_1, p_2, .., p_K]$ that exhibit high variability, but this variability is not properly tested when optimizing the detailed system to check for feasibility.

To limit this mismatch, we introduce the concept of ramping capability r_s, as a symmetric capability (up/down) for a given subsystem s. Equations (3.12) and (3.13) are added to the stage problem in (3.11).

$$p^H_{sk} - p^H_{s,k-1} - r_s \leq 0 \qquad \forall s, k \tag{3.12a}$$
$$p^H_{s,k-1} - p^H_{sk} - r_s \leq 0 \qquad \forall s, k \tag{3.12b}$$
$$0 \leq r_s \leq \overline{R}_s \qquad \forall s \tag{3.12c}$$

$$\gamma^v v + \gamma^e e + \gamma^r r + \kappa^v V_0 + \kappa^i I \leq \beta^F \tag{3.13}$$

The stage problem request subsystem ramping capability r_s that enters the feasibility check as a requirement R^*. It is not obvious how this requirement should be enforced by the detailed subsystem. As a starting point we require total production from the detailed system to ramp $r \geq R^*$ twice per day, in the morning (up) and in the evening (down). Note that this ramping capability should not be confused with the technical ramping capability of the system. The technical capability can be estimated and provided as the upper variable bound $\overline{R}_s$ in (3.12c). A request for high subsystem ramping capability will come in conflict with a request for high energy production since the ramping capability involves variations in the generation profile over the stage.

3.3.2 Reserve Capacity

A reserve capacity requirement can be included in the stage problem (2.7), where reserved capacity per subsystem (c_s) is accounted for in the generation limits. As a simplification, it is assumed that the area contributions are constant over the week and symmetric with respect to up- and down-regulation capacity.

Equations (3.14) are added to the stage problem in (3.11). Additionally, a reserve requirement should be included in the stage problem to define the need for reserve capacity. The feasibility cut can be extended as shown in (3.15).

$$p^H_{sk} - c_s \geq \underline{P}^H_s \qquad \forall s, k \tag{3.14a}$$
$$p^H_{sk} + c_s \leq \overline{P}^H_s \qquad \forall s, k \tag{3.14b}$$

$$\gamma^v v + \gamma^e e + \gamma^r r + \gamma^c c + \kappa^v V_0 + \kappa^i I \leq \beta^F \tag{3.15}$$

3.4 Feasibility Check

When checking hydropower subsystem feasibility, the aggregation and disaggregation of subsystem results and data are important.

3.4.1 Coupling Aggregated and Disaggregated Data

Fig. 3.4 illustrates how subsystem initial storage V_0 and subsystem inflow I are disaggregated to the detailed hydropower system, according to distribution factors ζ^V and ζ^I, correspondingly. The figure also illustrates how storage volumes in the detailed system can be aggregated to a subsystem storage v, according to distribution factors ζ^v.

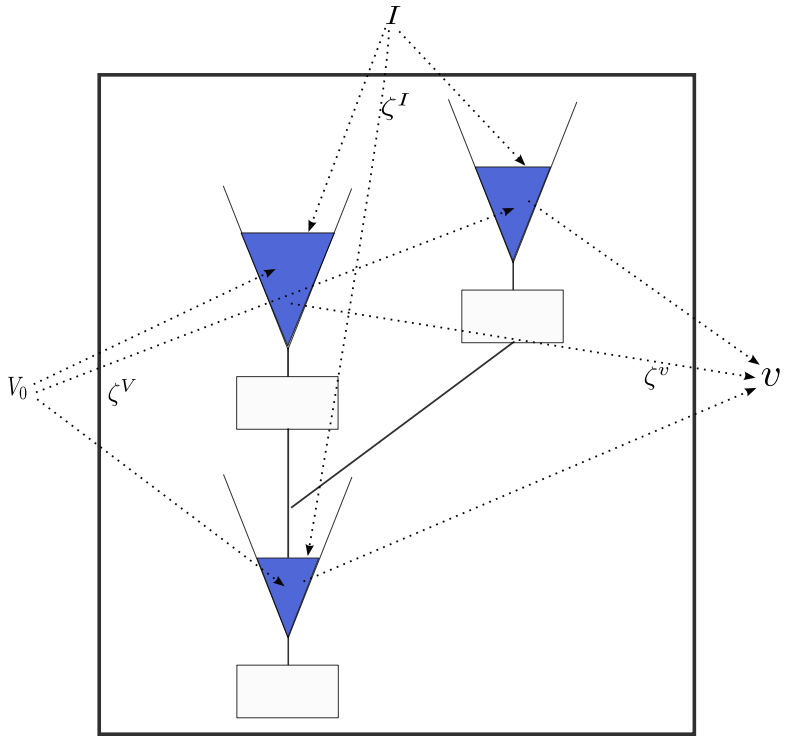


Figure 3.4: Aggregation and Disaggregation of hydropower data.

A simple way of defining reasonable distribution factors for inflow (both regulated and unregulated) is

$$\zeta_h^I = \frac{\tilde{I}_h}{\tilde{I}}, \tag{3.16}$$

where $\tilde{I}_h$ is the annual expected energy inflow to reservoir h, $\tilde{I}$ is the total amount of expected energy inflow for the subsystem. The expected inflow is provided as input, measured in water. Conversion from water to energy is done by using the best efficiency following the discharge waterway to sea. The vector ζ^I thus becomes the fractions of inflow to be received by each hydropower module in the feasibility check. Separate factors for regulated (ζ^R) and unregulated (ζ^U) inflow are computed.

3.4.2 Detailed Hydropower Optimization Problem

The feasibility check for a subsystem s for a stage t comprising time steps $k = 1, .., K$ is formulated in (3.17). Note that the formulation can be extended to include more technical details than what is presented here, but it should be formulated as an LP to provide dual information.

$$\mathcal{Z}_{st}(v_t^*, e_t^*, r_t^*, c_t^*, V_{t-1}^*, I_t^*) = \min\left(y^0 + y^v + y^e + y^r + y^c\right) \tag{3.17a}$$

$$V^0 \leq V_{t-1}^* + y^0 \qquad : \kappa^v \tag{3.17b}$$

$$I = I_t^* \qquad : \kappa^i \tag{3.17c}$$

$$\sum_{h\in\mathcal{H}} \eta_h v_{hK} + y^v \geq v_t^* \qquad : \gamma^v \qquad (3.17d)$$

$$\sum_{h\in\mathcal{H}} p_{hk}^H + y^e \geq e_t^* \qquad : \gamma^e \qquad (3.17e)$$

$$r + y^r \geq r_t^* \qquad : \gamma^r \qquad (3.17f)$$

$$c + y^c \geq c_t^* \qquad : \gamma^c \qquad (3.17g)$$

$$v_{hk} - v_{h,k-1} + \Gamma_w\left(q_{hk}^R + q_{hk}^S\right) - \Gamma_w\Big(\sum_{j\in\omega_h^D} q_{jk}^D + \sum_{j\in\omega_h^B} q_{jk}^B + \sum_{j\in\omega_h^S} q_{jk}^S\Big) = \zeta_k^V V^0 + \zeta_k^R I \qquad \forall h,k \qquad (3.17h)$$

$$q_{hk}^D + q_{hk}^B - q_{hk}^R = \zeta_k^U I \qquad \forall h,k \qquad (3.17i)$$

$$q_{hk}^D = \sum_{n\in\mathcal{N}_h} q_{nhk}^D \qquad \forall h,k \qquad (3.17j)$$

$$p_{hk}^H = \Gamma_e \sum_{n\in\mathcal{N}_h} \eta_{nh} q_{nhk}^D \qquad \forall h,k \qquad (3.17k)$$

$$\sum_{h\in\mathcal{H}} \left(p_{hk}^H - p_{h,k-1}^H\right) \geq r \qquad \forall h,k\in\mathcal{K}^+ \qquad (3.17l)$$

$$\sum_{h\in\mathcal{H}} \left(p_{h,k-1}^H - p_{hk}^H\right) \geq r \qquad \forall h,k\in\mathcal{K}^- \qquad (3.17m)$$

$$\sum_{h\in\mathcal{H}} p_{hk}^H + c \leq \overline{P}^H \qquad \forall k \qquad (3.17n)$$

$$\sum_{h\in\mathcal{H}} p_{hk}^H - c \geq \underline{P}_t^H \qquad \forall k \qquad (3.17o)$$

$$\underline{Q}_{ht}^D \leq q_{hk}^D \leq \overline{Q}_{ht}^D \qquad \forall h,k \qquad (3.17p)$$

$$\underline{Q}_{ht}^B \leq q_{hk}^B \leq \overline{Q}_{ht}^B \qquad \forall h,k \qquad (3.17q)$$

$$\underline{V}_{ht} \leq v_{hk} \leq \overline{V}_{ht} \qquad \forall h \qquad (3.17r)$$

The objective in (3.17a) is to minimize the use of slack variables y to ensure feasibility. The aggregation of detailed reservoir volumes v_{hK} at the end of the decision period to match energy content v_t^* is done in (3.17d), where η_h is the conversion rate (energy/water) according to best efficiencies following the discharge waterway to sea. The requirement in (3.17d) provides a flexible target, but may lead to unrealistic schedules for the detailed system. To compensate for that, measures for controlling completely full or empty reservoirs could be included. The energy (3.17e), ramping (3.17f), and reserve capacity (3.17g) targets are formulated next. Note that all these targets have an associated slack variable and that their dual variables γ are stored. The state variables representing initial energy storage (3.17b) and sum energy inflow (3.17c) have dual values κ.

The reservoir balances per hydropower storage h and time step k are presented in (3.17h). Distribution factors ζ^V and ζ^I are used for disaggregating initial energy storage and sum energy inflow to reservoirs, as explained above and illustrated in Fig. 3.4. Note that ζ_k^V is only non-zero for $k = 1$. Unregulated inflow is disaggregated according to ζ_k^U enters the power station of module h downstream its reservoir, as expressed in (3.17i).

The power production p_{hk}^H is expressed in (3.17k) as a concave piece-wise linear function of discharge, according to discharge segments n with unique efficiencies η_{nh}. Equation (3.17j) aligns discharge from the reservoir with use of the discharge segments.

The ramping capability r_t^* requested from the stage problem enters as a test for ability of the detailed system to ramp up in (3.17l) and down in (3.17m). The selection of time steps $\mathcal{K}^+$ for up-ramping and $\mathcal{K}^-$ can e.g. be done according to an assumed daily load profile, where one selects one time step per day for up-

ramping and one for down-ramping. The ability of the detailed system to procure reserve capacity c_t^* requested from the stage problem is formulated in (3.17n) and (3.17o). The capacity requirement is symmetric.

Slack variables $(y^0, y^v, y^e, y^r, y^c)$ are included to ensure feasibility of the LP problem itself. A particular challenge arises with seasonal-dependent minimum boundaries on discharge, bypass and reservoir volume. Note that a slack variable y^0 in (3.17b) is needed to ensure that the initial resource state is sufficient to meet the detailed requirements. Slack variables y^0 and y^v will in combination with careful tuning of ζ^V ensure that such water requirements can always be met.

3.4.3 Pumped Storage Plants

The formulation in (3.17) does not involve pumped-storage plants. These can be added by subtracting energy used for pumping from (3.17e). Consequently more generation capacity can be made available at the immediate cost of less production. This representation allows pumping motivated by both seasonal and short-term benefits. The seasonal aspect is taken care of when the subsystems end volume is increased on behalf of reduced net energy production. The short-term aspect can e.g. be motivated by delivering increased ramping capability.

3.4.4 Computing Feasibility Cuts

Once the LP problem in (3.17) is solved, one can determine whether the subsystem decisions $(v_t^*, e_t^*, r_t^*, c_t^*)$ are feasible provided the state (V_{t-1}^*, I_t^*). If the objective is zero, no slack variables y have been used and we can declare the subsystem decisions as feasible. On the contrary, if slack variables are used, we create a feasibility cut for the non-feasible subsystem operating point:

$$\begin{aligned}\gamma^v(v - v_t^*) + \gamma^e(e - e_t^*) + \gamma^r(r - r_t^*) + \gamma^c(c - c_t^*) \\ + \kappa^v(V_{t-1} - V_{t-1}^*) + \kappa^i(I - I_t^*) \le -Z^*\end{aligned} \tag{3.18}$$

Collecting parameters on the right-hand side so that:

$$\gamma^v v + \gamma^e e + \gamma^r r + \gamma^c c + \kappa^v V_{t-1} + \kappa^i I \le \beta^F, \tag{3.19}$$

where $\beta^F = -Z^* + \gamma^v v_t^* + \gamma^e e_t^* + \gamma^r r_t^* + \gamma^c c_t^* + \kappa^v V_{t-1}^* + \kappa^i I_t^*$

3.5 Visualization of Feasibility Spaces

With feasibility cuts of type (3.19) it is possible to visualize the resulting feasible polyhedron in 3D by discretizing three out of the four decision variables (v, e, r, c) and finding the binding cuts for each discrete combination. Fig. 3.5 shows a screen dump from a visualization program made by a Summer Researcher at SINTEF Energi in 2024. It shows feasibility spaces computed for a hydropower cascade under different conditions. Some visual patterns typically seen are discussed below:

- As energy production increases, the volume storage decreases. At high energy production output, the decrease is more intense due to lower efficiency at high power output. The concave PQ-curves are represented in the feasibility check in (3.1).
- As ramping capability increases, energy production decreases. This is explained by the fluctuating production profile required in the feasibility check in (3.1). Again, when energy production is close to maximum output, the efficiency is reduced.

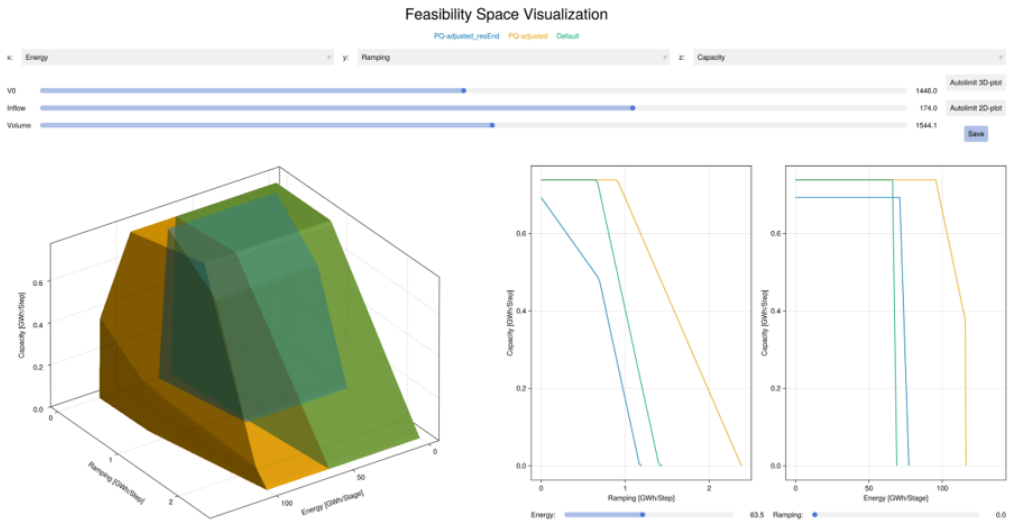


Figure 3.5: Feasibility space visualized.

3.6 Assumptions and Further Work

The use of feasibility spaces to approximate the detailed hydropower system relies on a series of assumptions:

- *Hydropower operation has no explicit cost components.* As discussed in the previous sections, the alternative cost of stored energy is accounted for in the SDDP algorithm, and the cost of operating pumps can be accounted for in (3.17).
- *Subsystem capabilities are expressed per stage.* As shown in [18] capabilities can be expressed per time step within the stage, but this modeling comes at the expense of significantly increased computational effort. Making subsystem decision per stage challenges the capability of representing intra-stage variations such as ramping. An approximate solution was presented in Section 3.3.1.
- *Disaggregation is approximated.* The aggregated system state, including the initial storage and inflow for the subsystem, needs to be disaggregated to the detailed problem in (3.17). Moreover, the aggregated energy storage at the end of the stage needs to be computed based on the detailed storage volumes. This topic was discussed in Section 3.4.1.

The work on feasibility spaces has been documented in [18] and [19], including mathematical descriptions and numerical analyses. Some possibilities for further work are listed below:

- Further testing and validation is needed. For example, one could test to what extent feasibility cuts may capture the impact of new constraints, planned revisions, pumped storage plants, etc.
- Improve the distribution of the subsystem's energy storage to the detailed hydro storages in the feasibility check. The process of balancing stoarge levels across complex hydropower storages is, to a large extent, controlled by the operator in real life. This process can borrow from concepts developed for the detailed drawdown part of the EMPS model [20].
- Improve the distribution of the subsystem's energy inflow to detailed hydro storages in the feasibility check. This process is uncertain and cannot be planned. One could explore feasibility spaces that are robust towards different and unfortunate distributions of inflow.
- Further study and possibly improve on the aggregate capabilities of hydropower representation on subsystem level.

Chapter 4

Residual Demand Cost Function

With the high variability in the power system caused by increasing amounts of wind and solar power, there is a growing need for fine time resolution in scheduling models. As this variability will impact the economic dispatch, and consequently the marginal cost of supplying demand, it should also be considered when valuating water. With modeling at a finer time resolution comes higher computation times, and thus a need for careful modeling.

So far we have treated the stage decision problem as an LP according to (3.11) where the hydropower system is approximated by feasibility cuts as described in Chapter 3. Hydropower subsystems will mainly allow flexible operation within a decision stage (week). Recall that their storages are state variables in the stochastic dynamic programming scheme, and we can therefore assume that they most of the time can accommodate variations within the week without reaching their storage boundaries. This leads to the idea of optimizing the use of hydropower for the whole week against a *residual demand cost function* (RDCF) which represents 'everything else' being part of the economic dispatch problem. The separation between hydropower system itself and the RDCF for hydro is done by use of two-stage Benders decomposition, serving to valuate hydropower capabilities.

In this work we follow the decomposition principle first discussed in [21] and later explored in [22] and [23]. The concept is illustrated in Fig. 4.1, indicating abstraction of the residual demand for hydro by means of decomposition. The decomposition allows us to represent the RDCF as a function of the hydropower decision variables, expressed as linear inequalities, or *cost cuts*, as explained in the following.

4.1 Derivation of the Residual Demand Cost Function

The idea is to express the RDCF ϕ of operating the system in a given stage as a function of the sum hydropower production for the subsystems (e_s). This cost function is illustrated in one dimension in Fig. 4.2. The "true" cost of operating the system is reduced with higher hydropower production according to a blue curve in the figure. The convexity follows from the linear formulation of the complete problem in (3.11).

We define a cost function ϕ where the objective and constraints comprise the stage problem as formulated in (3.11), except from the hydropower representation. The vector of hydropower productions e per stage enters the problem in (4.1) as requirements and are freely divided per time step so that $\sum_{k\in\mathcal{K}} p^H_{sk} = e_s$ for $s \in \mathcal{S}$.

$$\phi(e) = \min c^\intercal x, \qquad Ax \geq b(e) \quad : \mu \tag{4.1}$$

This function can be described by Benders cuts, as illustrated in Fig. 4.2. We will refer to these cuts as *cost cuts* in the following.

$$\phi + \sum_{s\in\mathcal{S}} \mu_s \left(e_s - e^*_s\right) \geq \phi^* \tag{4.2}$$

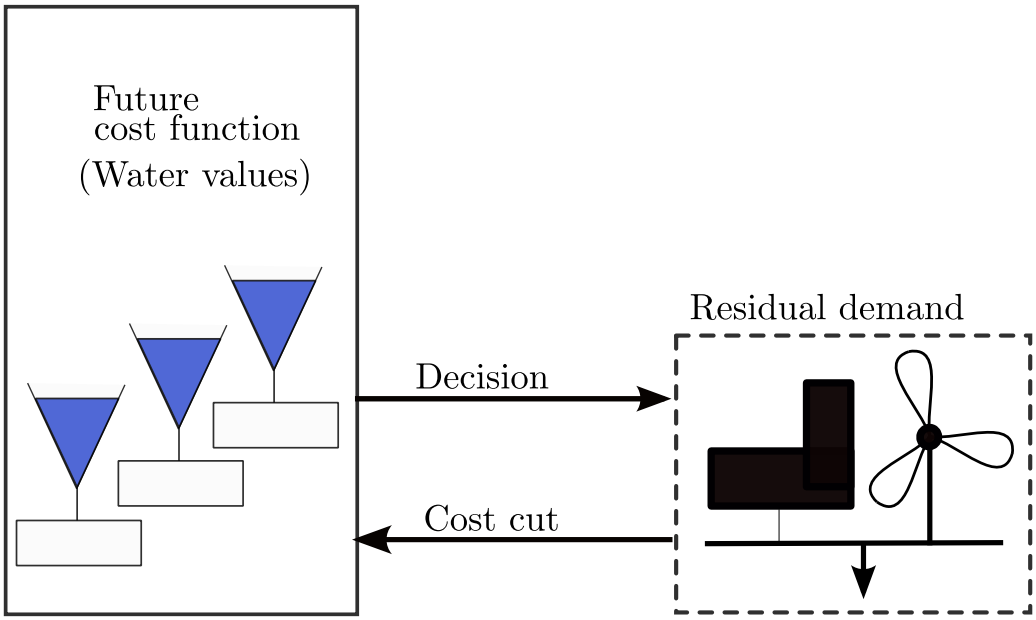


Figure 4.1: Illustration of decomposition between the hydropower system and its residual demand.

The stage problem can be further simplified from (3.11) in (4.3).

$$\mathcal{Q}_t(\mathbf{v}^*_{t-1}, \mathbf{z}_{t-1}) = \min\Big(\phi_t + \alpha_{t+1}\Big) \tag{4.3a}$$

$$s.t \qquad \text{(3.11b)} - \text{(3.11e)}$$

$$v_{st} + e_{st} + s_{st} = v_{s,t-1} + I_{st} \qquad \forall s \tag{4.3b}$$

$$\gamma^v_{ct} v_{st} + \gamma^e_{ct} e_{st} \leq \beta^F_{ct} \qquad \forall s, c \tag{4.3c}$$

$$\phi_t + \sum_{s \in \mathcal{S}} \mu^e_{sct} e_{st} \geq \beta^C_{ct} \qquad \forall c \tag{4.3d}$$

$$\alpha_{t+1} + \sum_{s \in \mathcal{S}} \pi^v_{sct} v_{st} + \sum_{s \in \mathcal{S}} \pi^z_{sct} z_{st} \geq \beta_{ct} \qquad \forall c \tag{4.3e}$$

$$\underline{E}_{st} \leq e_{st} \leq \overline{E}_{st} \qquad \forall s \tag{4.3f}$$

The problem in (4.3) seeks the volume and production that minimizes the sum of the RDCF (ϕ) and the FCF (α) while respecting the weekly reservoir balances and the feasibility cuts. Note the lightweight form of the stage problem in (4.3) compared to the original formulation in (2.7). In summary, the new stage problem represents:

- Hydropower per subsystem by feasibility cuts in (4.3c) expressed in terms of energy storage v and energy production e for that subsystem.
- The residual demand for hydro by cost cuts in (4.3d) expressed in terms of energy production e for all subsystems.
- The future value of stored energy in hydropower subsystems by FCF cuts in (4.3e) expressed in terms of energy storage v.

Since equations (4.3c), (4.3d), and (4.3e) are all linear inequalities of relatively low dimensionality, but possibly with a high number of cuts in each set, proper cut management may be a powerful technique for reducing computation time when solving instances of (4.3).

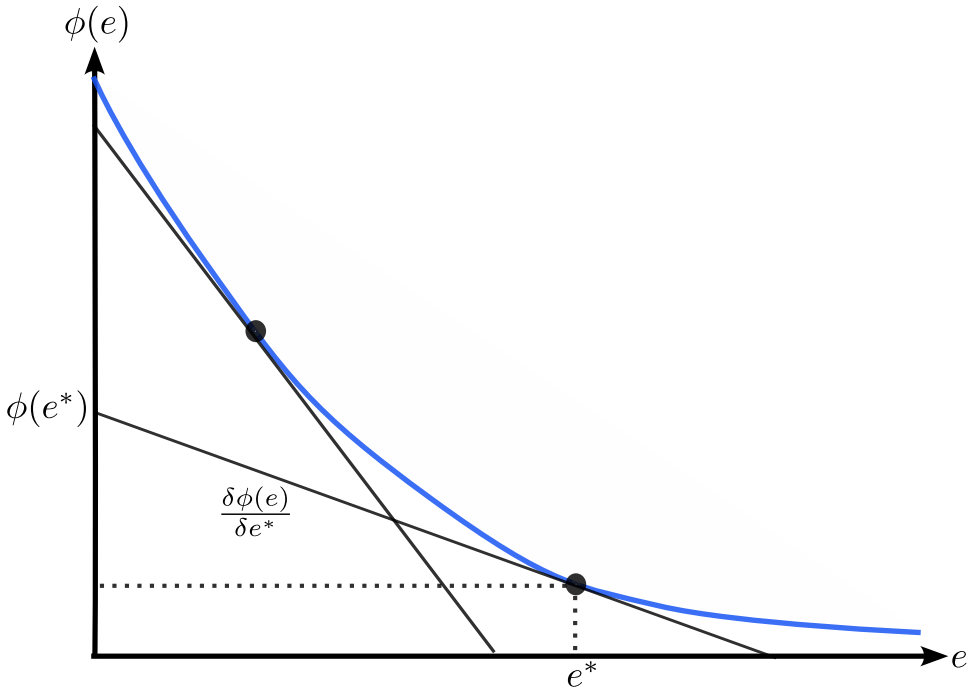


Figure 4.2: Illustration of residual demand cost function and creation of cuts.

4.2 Graphical Interpretation

Fig. 4.3 illustrates the FCF and RDCF functions in the dimension of one state. Note that the e axis is reverted compared to Fig. 4.2 for illustration purposes. The stapled gray line shows the total cost as a function of energy storage (v) and production (e). Typically, the RDCF will not span the entire volume since it will not be possible to use the maximum volume for energy production within one single stage. Thus, the blue curve in Fig. 4.3 should bee seen as an illustration only. Moreover, we make the assumption that the production of one unit e corresponds to the loss of one unit of storage v, disregarding the impact of spillage and bypass.

The optimal production will minimize the sum of immediate and expected future costs. This optimum can be found by evaluating the KKT optimality conditions. In the case of one hydro subsystem and no binding constraints:

$$\frac{\partial \mathcal{Q}_t}{\partial e_t} = 0 \rightarrow \frac{\partial \phi_t}{\partial e_t} + \frac{\partial \alpha_{t+1}}{\partial e_t} = 0 \tag{4.4}$$

Reformulating and exploiting that $\frac{\partial v_t}{\partial e_t} = -1$ eventually leads to (4.6).

$$\frac{\partial \phi_t}{\partial e_t} + \frac{\partial \alpha_{t+1}}{\partial v_t} \times \frac{\partial v_t}{\partial e_t} = 0 \tag{4.5}$$

$$\frac{\partial \phi_t}{\partial e_t} = \frac{\partial \alpha_{t+1}}{\partial v_t} \tag{4.6}$$

We seek a decision where the marginal residual demand for hydropower equals the expected marginal cost of storing water, as illustrated in Fig. 4.4. This decision leads to the water value $\frac{\partial \alpha_t}{\partial e_t}(x^*)$ at the beginning of the decision period for the state being considered (unless other boundaries or constraints are faced). Note that this is the main principle used in traditional algorithms for water value calculations used in the EMPS model [24]: One seeks to 'operate along a constant water value'.

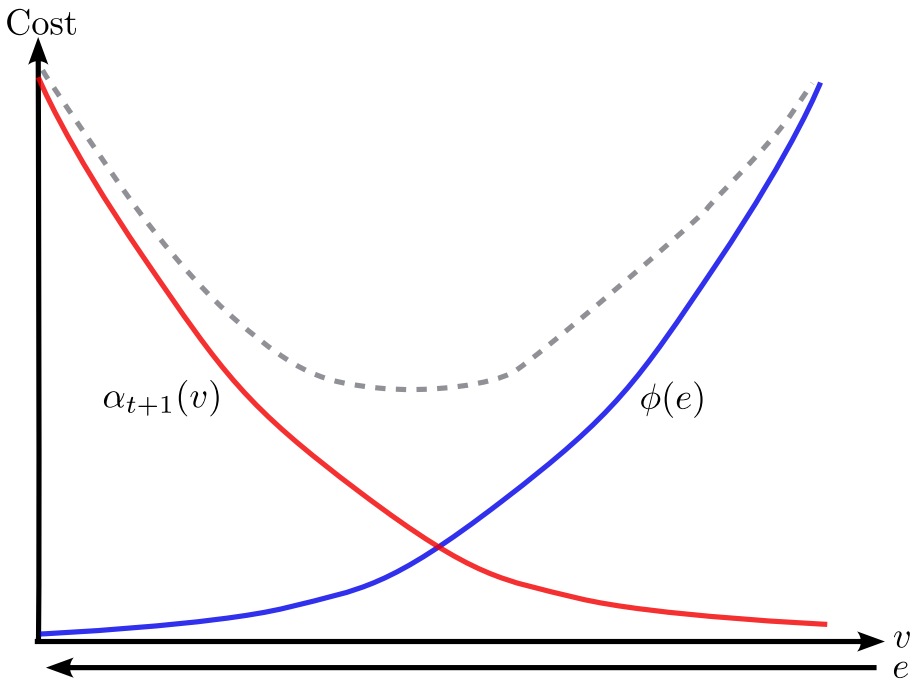


Figure 4.3: Illustration of how the optimal use of water is found by use of the two cost functions.

4.3 Improved Formulation

With the formulation in (4.3), the sum of hydropower generation per stage can be freely distributed to time steps in the detailed problem in (4.8). Consequently, we run the risk of overestimating the capability of the hydropower subsystem to adapt to rapid variations in net demand in the residual market problem.

It is possible to constrain operation of hydropower by use of more detailed feasibility cuts where the capabilities of the hydropower subsystems are expressed in terms of decision variables (v, e, r, c). The cost cuts from (4.3d) can then be expanded to (4.7). Since $\frac{\delta\phi}{\delta e}$, $\frac{\delta\phi}{\delta r}$, $\frac{\delta\phi}{\delta c}$ are all ≤ 0, high values of e, r, c will be sought in (4.7) to seek a low immediate cost. While the production e has a direct relationship with the FCF through the reservoir balance in (4.3b), the ramping capability r and reserve capacity c are constrained only through feasibility cuts in (4.3c).

$$\phi_t + \sum_{s\in\mathcal{S}} \mu^e_{sct} e_{st} + \sum_{s\in\mathcal{S}} \mu^r_{sct} r_{st} + \sum_{s\in\mathcal{S}} \mu^c_{sct} c_{st} \geq \beta^C_{ct} \qquad \forall c \tag{4.7}$$

A basic version of the residual market problem solved to find the cut parameters in (4.7) is expressed in (4.8). Note that additional features that can be expressed as linear or piece-wise linear functions are easily added to this problem without compromising the properties of the decomposition scheme.

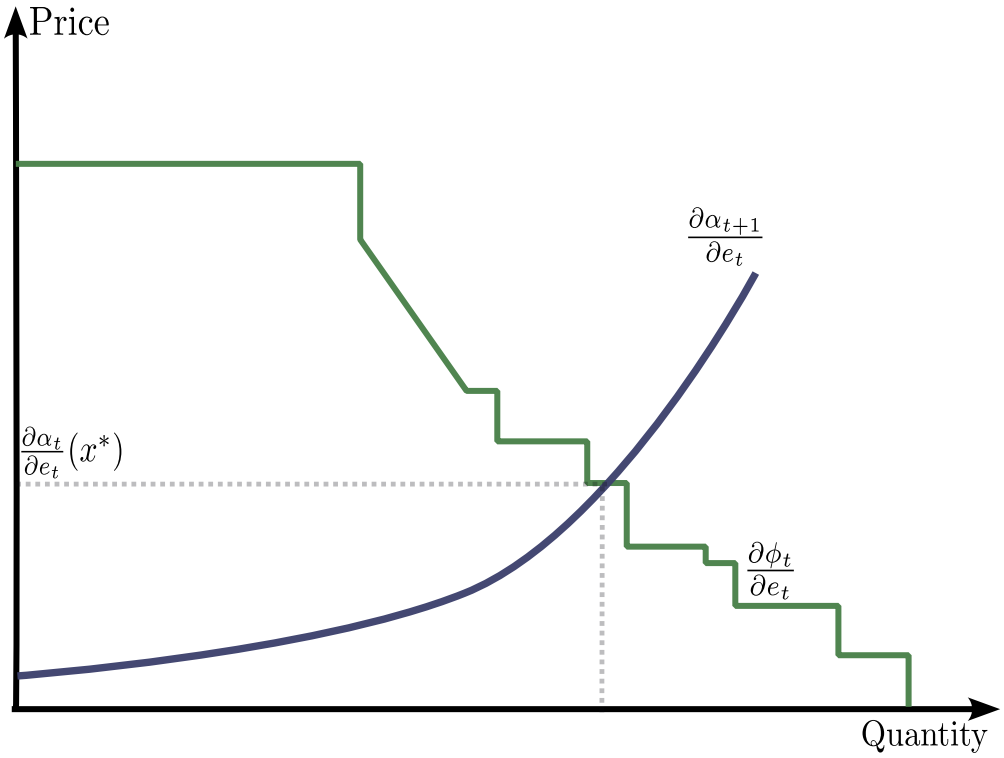


Figure 4.4

$$\phi(e_s^*, r_s^*, c_s^*) =$$

$$\min \sum_{k\in\mathcal{K}} \sum_{a\in\mathcal{A}} \left[\sum_{g\in\mathcal{G}_a} C_{gt}^G p_{gk}^G - \sum_{d\in\mathcal{D}_a} C_{dt}^D p_{dk}^D + C^R p_{ak}^R \right] \tag{4.8a}$$

$$\sum_{k\in\mathcal{K}} p_{sk}^H = e_s^* \qquad : \mu_s^e \qquad \forall s \tag{4.8b}$$

$$r_s = r_s^* \qquad : \mu_s^r \qquad \forall s \tag{4.8c}$$

$$c_s = c_s^* \qquad : \mu_s^c \qquad \forall s \tag{4.8d}$$

$$\sum_{h\in\mathcal{H}_a} p_{hk}^H + \sum_{g\in\mathcal{G}_a} p_{gk}^G - \sum_{d\in\mathcal{D}_a} p_{dk}^D + p_{at}^R + \sum_{\ell\in\mathcal{L}_a^+} f_{\ell k} - \sum_{\ell\in\mathcal{L}_a^-} f_{\ell k} \geq D_{ak} - W_{ak} \qquad \forall a, k \tag{4.8e}$$

$$-r_s \leq p_{sk}^H - p_{s,k-1}^H \leq r_s \qquad \forall s, k \tag{4.8f}$$

$$\underline{P}_s^H \leq p_{sk}^H - c_s \qquad \forall s, k \tag{4.8g}$$

$$p_{sk}^H + c_s \leq P_s^H \qquad \forall s, k \tag{4.8h}$$

$$0 \leq p_{dk}^D \leq \overline{P}_{dk}^D \qquad \forall d, k \tag{4.8i}$$

$$0 \leq p_{ak}^R \leq \overline{D}_{ak} \qquad \forall a, k \tag{4.8j}$$

$$-\overline{F}_{\ell t} \leq f_{\ell tk} \leq \overline{F}_{\ell t} \qquad \forall \ell, k \tag{4.8k}$$

4.4 Computational Tests

This section report on basic numerical tests that were conducted for the purpose of understanding and building trust in the decomposition technique described so far in this chapter. From the two examples provided below, we can see that as the residual demand cost issue becomes more complex to account for interconnected systems and detailed time resolution, there is an opportunity for enhanced computational performance.

4.4.1 Example 1

To demonstrate the use of cost cuts, we established a test system comprising one hydropower reservoir and station, several thermal generators and a fixed demand, as illustrated in Fig. 4.5. All units are in 'energy' or 'money/energy'. We define a scheduling horizon of 8 stages and with uncertainty in inflow. The inflow may take two equally probable values for each stage, either 5 or 15. This leads to a scenario tree comprising $2^8 - 1 = 255$ nodes and $2^{8-1} = 128$ scenarios. Demand has a fixed value of 20 per stage, and maximum hydropower storage and production capacities are both 30. 20 thermal generators are included, each with maximum generation capacity of 1 and gradually increasing marginal costs $\{1.0, 2.0, ..., 20.0\}$. We included spillage and curtailment variables to ensure robustness, although not needed in our experiments. An initial reservoir volume of 5 and a first stage inflow of 10 are assumed.

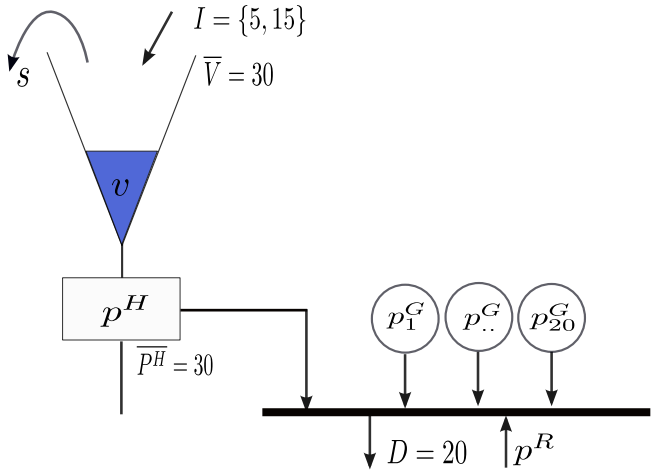


Figure 4.5: Test system

Solving this problem as an extensive form stochastic program gives the objective of 429.3125. The problem has 24 variables (20 thermal p^G, curtailment p^R, reservoir v, discharge p^H, and spillage s) and 2 constraints (reservoir and power balance) per node, giving a total of 6120 variables and 510 constraints.

Next we solve the stochastic program using two versions of the Benders decomposition algorithm. The first one, which we refer to as *Standard*, uses the classical multi-stage Benders decomposition algorithm, see e.g. [3]. For each node, the following LP problem is solved both in the forward and backward iterations:

$$\min \sum_{i=1}^{20} c_i p_i^G + \alpha \tag{4.9a}$$
$$v + p^H + s = v^0 + I \tag{4.9b}$$
$$p^H + \sum_{i=1}^{20} p_i^G + p^R = D \tag{4.9c}$$
$$\alpha + \pi_c v \geq \beta_c \qquad \forall c \in \mathcal{C}_1 \tag{4.9d}$$
$$0 \leq p_i^G \leq 1 \tag{4.9e}$$
$$0 \leq p^H \leq \overline{P^H} \tag{4.9f}$$
$$0 \leq v \leq \overline{V} \tag{4.9g}$$

We refer to (4.9) as the *node problem*. It has 25 variables (including α) and 2 basic constraints (reservoir and power balance). Benders cuts of type (4.9d) are added in the backward recursion. These cuts are shared among all nodes in the same stage.

In the second decomposition version, which we refer to as *CostApprox*, the residual market is approximated using cost cuts of type (4.10d) that are generated in the forward iteration. For each node visited in the forward iteration, the node problem is divided into a *reduced node problem* in (4.10) and a market problem in (4.11).

$$\min \phi + \alpha \tag{4.10a}$$
$$v + p^H + s = v^0 + I \tag{4.10b}$$
$$\alpha + \pi_c v \geq \beta_c \qquad \forall c \in \mathcal{B} \tag{4.10c}$$
$$\phi + \mu_c p^H \geq \beta_c^C \qquad \forall c \in \mathcal{C} \tag{4.10d}$$
$$0 \leq p^H \leq \overline{P^H} \tag{4.10e}$$
$$0 \leq v \leq \overline{V} \tag{4.10f}$$

The reduced node problem has 5 variables and one basic constraint (reservoir balance). In addition, value cuts in (4.10c) and cost cuts in (4.10d) are added in the backward recursion and forward simulation, correspondingly. The cost cuts are generated by solving the market problem in (4.11) for realizations of p^{H*}.

$$\min \sum_{i=1}^{20} c_i p_i^G \tag{4.11a}$$
$$p^H = P^{H*} \quad : \mu \tag{4.11b}$$
$$p^H + \sum_{i=1}^{20} p_i^G + p^R = D \tag{4.11c}$$
$$0 \leq p_i^G \leq 1 \tag{4.11d}$$

As will be further discussed in Chapter 5, the creation of cost cuts comes as a natural addition to the forward iteration. The iterative procedure between (4.10) and (4.11) – where the former creates sampled values of p^{H*} and the latter creates corresponding cost cuts (4.10d) – can be carried out until convergence. However, this is not necessary from a mathematical point of view since we are approximating the objective in (4.9) from below.

The convergence of the Standard and CostApprox decomposition algorithms are shown in Fig. 4.6. CostApprox-1 refers to the computation of only one cost cut per node in the forward iterations, while CostApprox-N refers to the process of iterating between (4.10) and (4.11) until convergence. All three decomposition approaches converge to the same optimal objective as the full stochastic program (429.3125) in 11 iterations.

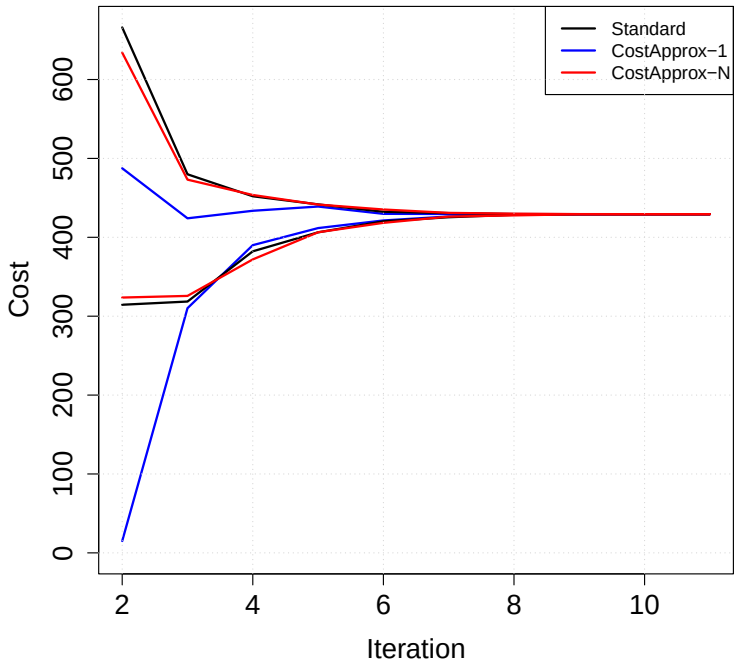


Figure 4.6: Convergence for Benders decomposition algorithms.

Some interesting observations can be made from the convergence plots in Fig. 4.6. Firstly, the Standard approach (black curve) obtains a tighter lower bound in the early iterations than CostApprox-1 (blue curve). All lower bounds will increase with iterations since we are gradually constraining the problem, but for CostApprox-1 this process is clearly slower than for the two other approaches. Secondly, the upper bound of CostApprox-1 increases between iterations 3-5. This increase is due to the fact that we in the forward iteration are not solving the node LP problems to optimality, and thus underestimate the costs in the early iterations.

With the CostApprox algorithm, we seek to balance the RDCF and FCF functions in each node, as described in (4.10). The RDCF decreases as we produce more hydropower (or decrease reservoir volume). The FCF decreases as we store more energy for later stages. All cuts computed for stages 2-3 and 4-5 are shown in Fig. 4.7 and Fig. 4.8, respectively. The x-axis is treated as energy storage for the future cost cuts (blue) and for production (starting at full storage and no production). To find the balance points that minimize sum costs one would have to know the initial reservoir and inflow for the node. Note that the cost cut gradients reflect the marginal costs of various thermal generators in this case, while the steepest red line reflects the marginal cost of curtailment.

From Figs. 4.7 and 4.8 it is clear that some of the FCF cuts are redundant, in the sense that they are dominated by other cuts over the whole solution domain, which is due to the multi-stage nature of the problem. In contrast, the cost cuts are found as part of a two-stage decomposition approach and are thus always binding in their corresponding operating point.

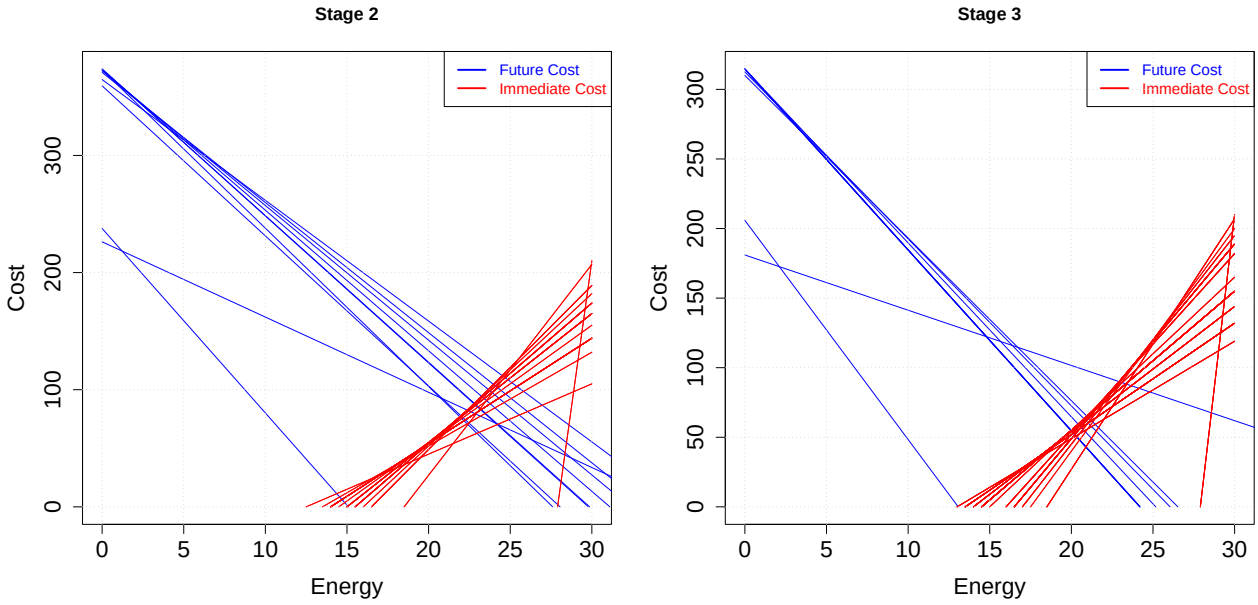


Figure 4.7: Cuts of type (4.10d) (red) and (4.10c) (blue) for stages 2 and 3.

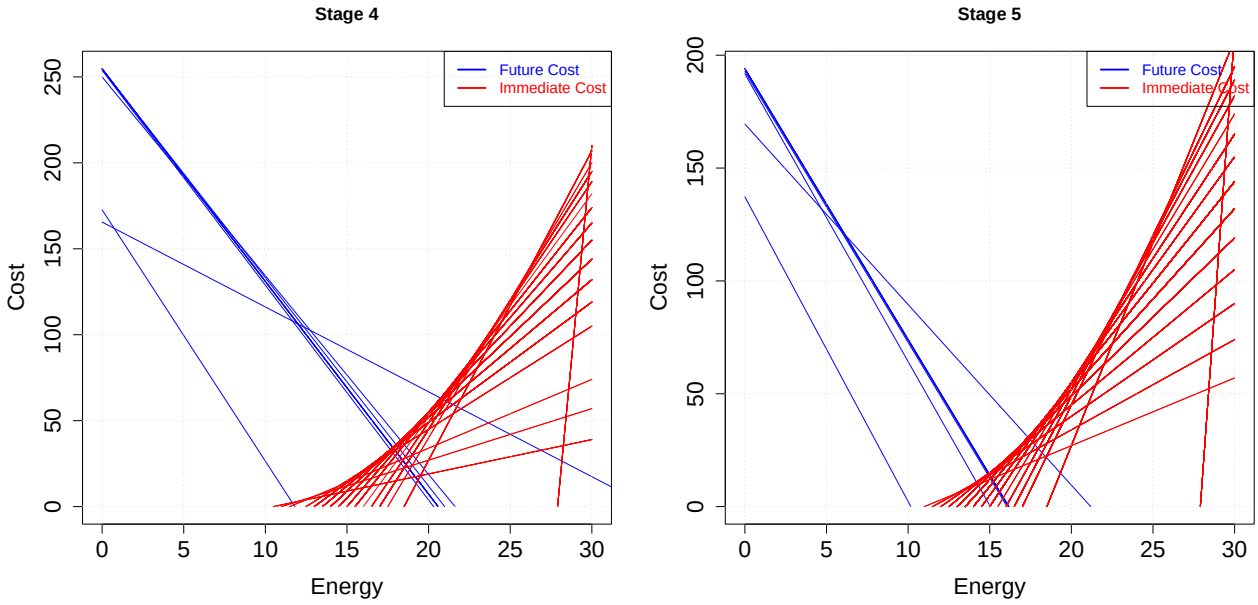


Figure 4.8: Cuts of type (4.10d) (red) and (4.10c) (blue) for stages 4 and 5.

4.4.2 Example 2

We expand the problem in Example 1 to comprise 168 time steps within each of the 8 decision stages. Generation capacities and loads are scaled according to the shorter decision stages, and a randomness is introduced to ensure load variation within the decision periods. The CostApprox-1 takes 57 seconds to solve while the Standard decomposition approach which spends 194 seconds. Hardware specifications are stated in Chapter 6. The convergence plots are presented in Fig. 4.9.

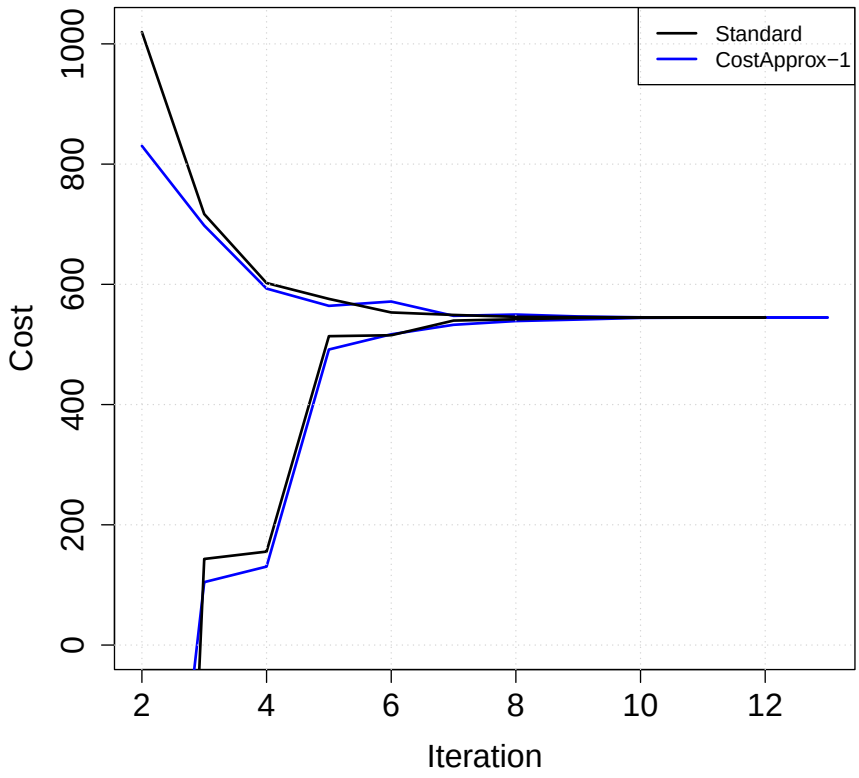


Figure 4.9: Convergence for Benders decomposition algorithms.

There are two notable differences in computational complexity between Standard and CostApprox-1 which serve to explain the large difference in computation time:

1. In CostApprox, the residual market is only explicitly modeled in the forward iteration, to create cost cuts. The cost cuts are expressed in terms of sum hydro production per decision stage (and not per time step), and these cuts represent the residual market in the backward recursion. Thus, the backward recursion is substantially faster in CostApprox than in Standard.

2. The intra-stage reservoir balances are ignored in CostApprox. This is a simplification which is reasonable for larger reservoirs. For cases where the intra-stage reservoir balances are temporarily violated, the CostApprox thus overestimates the storage capability. As a test, the maximum reservoir capacity in Example 2 was reduced to 20, giving a higher cost for Standard than for CostApprox. Moreover, spillage was detected in Standard, but not in CostApprox.

4.5 Short-Term Uncertainties

The presentation so far has focused on representing uncertainty in inflow on a weekly level, according to an VAR1 model. Inflow is often considered the major source of uncertainty in hydro-dominated systems. It typically show a significant autocorrelation between weeks, necessitating inflows as state variables in SDDP. Moreover, inflow predictions within a decision stage (week) are considered to be appropriately accurate for our purpose.

Several other sources of uncertainty may be relevant to include in the scheduling. In this work we are particularly interested in representing the uncertainty in VRES (wind and solar power). Unlike for inflow, the variability in VRES is typically more difficult to predict for the near future, while longer term auto-correlations are muted. Thus, we argue that VRES uncertainty can be characterized as short-term, and wind and solar power can be neglected as state variables in the SDDP algorithm. Further statistical analyses should be done to support such statements for the system at hand, see e.g. [13] and [14] for related discussions.

Creation of cost cuts naturally fits in to the forward simulation phase, where the state trajectories are simulated according to sampled inflow scenarios. The process of creating cost cuts is illustrated in Fig. 4.10. First, **StageProb**, formulated as (5.1), is solved. Subsequently, the hydropower production is passed to the residual market problem **CostProb**, formulated as (4.8). Along the lines of [23], short-term uncertainties can be revealed in the second stage, solving CostProb for NW different realizations of VRES. A total of NC iterations between LP-FWD and LP-COST are performed.

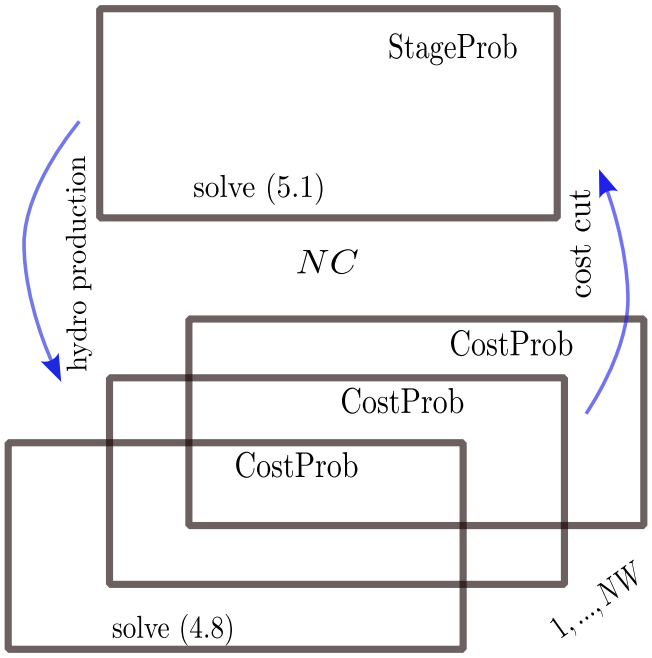


Figure 4.10: LP problems solved in forward iterations.

Cost cuts can either be generated for each of the NW scenarios (multi-cut) or as an aggregate cut where cut coefficients are probability-weighted. In the following we use the aggrgated cuts.
The outlined approach for representing short-term uncertainties are based on two assumptions:

- The hydropower productions are part of the first-stage decision. Thus, these decisions are made with information about VRES uncertainty, but are not tailored to a specific VRES scenario.
- VRES are not SDDP state variables. Thus, VRES realizations in a given decision stage do not depend on realizations in the previous stage. Moreover, it is assumed that wind and inflow are uncorrelated.

Under the assumptions above, cost cuts can be shared among different states in a given stage without further adjustments. Note that the two-stage stochastic approach allows for other types uncertainties to be represented, as long as the two assumptions above can be made. Uncertain parameters entering the objective (such as prices) and the constraint matrix (such as topological changes) may therefore be considered.

The treatment of VRES uncertainty within the ReSDDP model is also described in [25], supported by numerical examples.

4.6 Assumptions and Further Work

The approximation of the residual demand for hydropower within long-term hydropower scheduling models by means of cost cuts is based on a set of assumptions:

- *Hydropower operation can be optimized at coarse time resolution.* Through this assumption we argue that hydropower operation is flexible and sees no (or few) operational constraints within the week. While this may be true for large and aggregated systems, many intra-week constraints exists and may be important for the formation of water values.
- *Hydropower decisions are made before knowing short-term uncertainties.* Following the representation of short-term uncertainties described in Section 4.5, weekly decisions for the hydropower system are made before knowing weekly VRES outcomes. This allows the cost cuts to embed short-term uncertainty. The VRES realizations are known for the whole week at the time.
- *The residual demand optimization problem is convex.* The least-cost dispatch problem for the week should be formulated as a convex optimization problem to allow for computation of cost cuts.

The work on approximations of the residual demand for hydro has been documented in [21–23, 25], including mathematical descriptions and numerical analyses. Some possibilities for further work are listed below:

- Further investigating the representation of hydropower system. While the modeling of hydropower decisions at a coarse time resolution is important for the computational performance, the need for aggregation and the possibility of further constraining of its flexibility in the market problem should be investigated.
- Further investigating the representation of the residual market problem. The majority of further experimentation with the ReSDDP model within the RES100 project will directly concern the addition of new functionality in the market problem. New features such as demand response, batteries, and hydrogen will be implemented and studied. Other aspects, such as detailed grid constraints, (linearised) unit commitment, and ramping constraints may also be included.
- Offline approximation of the residual demand function. The proposed scheme evaluates the market problem in the forward SDDP iterations. This allows the residual demand function to be evaluated around obtained operating points. Optionally, one could experiment with a priori estimation of this function.
- Representation of short-term uncertainties. Studies to further investigate the impact of short-term uncertainty should be carried out. Different uncertainty structures and types could be investigated.

Chapter 5

ReSDDP

So far we have defined the long-term scheduling problem in Chapter 2, and described how decomposition by feasibility and cost cuts can be applied to isolate the computation of future cost functions for the hydropower subsystems, in Chapters 3 and 4, respectively. The future cost functions are computed in the SDDP algorithm. Both the feasibility and the cost cuts can be seen as spatial decomposition techniques, serving to reduce the dimensionality of the stage problem solved in the SDDP algorithm.

The resulting scheduling model will provide strategies, in the form of future expected cost functions per decision stage. These strategies are valid for the long-term storage capability provided by hydropower subsystems and can be used in more detailed simulation models. A desirable property of the outlined decomposition scheme is that it allows explicit representation of all detailed technologies and power system features into the computation of the FCFs:

- Hydropower details are embedded in the fasibility cuts
- Other technologies on supply and demand side as well as grid bottlenecks are reflected in the cost cuts

This allows quantifying the impact of various features on the water values in a transparent manner, where the accuracy of the approximations can be controlled to a fine level.

The ReSDDP is modeled in Julia using the optimization package JuMP [26]. The computer code is maintained at the Gitlab repository at `https://gitlab.sintef.no/energy/res100/`.

5.1 Approximated Stage Problem

As discussed in Chapter 2, the SDDP algorithm repeatedly solves a stage problem in a forward simulation and backward recursion. The detailed stage problem was described in Chapter 2 as (2.7). With the introduction of feasibility cuts and cost cuts the detailed stage can be approximated as:

$$\mathcal{Q}_t(\mathbf{v}^*_{t-1}, \mathbf{z}_{t-1}) = \min\left(\phi_t + \alpha_{t+1}\right) \tag{5.1a}$$

$$\mathbf{v}_{t-1} = \mathbf{v}^*_{t-1} \quad : \boldsymbol{\pi}^{\upsilon} \tag{5.1b}$$

$$\mathbf{z}_{t-1} = \mathbf{z}^*_{t-1} \quad : \boldsymbol{\pi}^{z} \tag{5.1c}$$

$$\mathbf{z}_t = \boldsymbol{\Phi}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t \tag{5.1d}$$

$$I_{st} = \sigma_{ht} z_{st} + \mu_{st} \quad \forall s \tag{5.1e}$$

$$\upsilon_{st} + e_{st} + s_{st} = \upsilon_{s,t-1} + I_{st} \quad \forall s \tag{5.1f}$$

$$\gamma^{\upsilon}_{sct}\upsilon_{st} + \gamma^{e}_{sct}e_{st} + \gamma^{r}_{sct}r_{st} + \gamma^{c}_{sct}c_{st} + \kappa^{\upsilon}_{sct}V_{s,t-1} + \kappa^{i}_{sct}I_{st} \leq \beta^{F}_{sct} \quad \forall s, \forall c \in \mathcal{F} \tag{5.1g}$$

$$\phi_t + \sum_{s\in\mathcal{S}} \mu^{e}_{sct}e_{st} + \sum_{s\in\mathcal{S}} \mu^{r}_{sct}r_{st} + \sum_{s\in\mathcal{S}} \mu^{c}_{sct}c_{st} \geq \beta^{C}_{ct} \quad \forall c \in \mathcal{C} \tag{5.1h}$$

$$\alpha_{t+1} + \sum_{s\in\mathcal{S}} \pi^{\upsilon}_{sct}\upsilon_{st} + \sum_{s\in\mathcal{S}} \pi^{z}_{sct}z_{st} \geq \beta_{ct} \quad \forall c \in \mathcal{B} \tag{5.1i}$$

$$0 \leq \upsilon_{st} \leq \overline{V}_{st} \quad \forall s \tag{5.1j}$$

$$\underline{E}_{st} \leq e_{st} \leq \overline{E}_{st} \quad \forall s \tag{5.1k}$$

$$0 \leq r_{st} \leq \overline{R}_{st} \quad \forall s \tag{5.1l}$$

$$0 \leq c_{st} \leq \overline{C}_{st} \quad \forall s \tag{5.1m}$$

The first two constraints, (5.1b) and (5.1c), serve the purpose of finding the dual values for the state variables storage (π^{υ}) and inflow (π^{z}), respectively. The next two, (5.1d) and (5.1e), displays the computation of inflow that enters the reservoir balance in (5.1f). The feasibility cuts derived in Chapter 3 are described in (5.1g). The cost cuts derived in Chapter 4 are described in (5.1h). FCF cuts describing the future expected cost function are presented in (5.1i). Finally, variable boundaries are stated in (5.1j)-(5.1m).

The problem formulated in (5.1) has few variables since decisions are aggregated both in space and time. That is, decisions are made per hydropower subsystem (as opposed to per module) and for the entire decision stage t (as opposed to per time step k). The major computational complexity is associated with cuts in (5.1g)-(5.1i). All cut types serve to approximate the stage problem, and the quality of the approximation can be controlled as elaborated in [18] for feasibility cuts and [23] for cost cuts. Generally, a fine approximation requires a high number cuts, challenging the computational complexity of (5.1). Finding the appropriate number of cuts in sets $\mathcal{F}$, $\mathcal{C}$, and $\mathcal{B}$ to properly approximate the respective functions becomes an important challenge.

5.2 Cut Management

FCF cuts of type (5.1i) are computed in the SDDP backward recursion. These cuts are not necessarily tight since we have decomposed the overall problem into multiple stages. Efficient treatment of Benders cuts within the SDDP algorithm is important to achieve reasonable computation times for large-scale problems. Several techniques are suggested in the literature, see e.g. [27]. A well-known technique is often referred to as relaxation (or dynamic cut selection in [27]). It involves: (i) solving the decision problem without any cuts; (ii) checking if cuts are violated by inserting trial decisions; and (iii) adding the most violated cut(s). Such relaxation techniques can be very efficiently implemented in compiled programming languages, but at the time of writing we did not see a significant speed-up when implementing relaxation in our Julia/JuMP code. Thus, we do not consider cut relaxation for neither of the cut types in this work. Another technique is to remove redundant cut according to the elimination procedure outlined in [28] (Section 4.3).

The feasibility cuts in (5.1g) and cost cuts in (5.1h) are derived from two-stage Benders decomposition approaches and are therefore always tight in the states for which they were created. In other words, and unlike the case for FCF cuts in a multi-stage decomposition procedure, creating the cut once for a given state is sufficient. This opens for the possibility of creating the feasibility and cost cuts before running the SDDP algorithm.

As an example, the creation of feasibility cuts was made both prior to running SDDP (static) or as part of the forward simulation (dynamic) in [18]. The static approach requires either an enumeration of the state space or a qualified best guess of which states that are likely to be visited in the SDDP run. The ReSDDP implementation currently allows feasibility cuts to be computed on beforehand by use of enumeration. Since feasibility cuts are computed per subsystem in isolation, the state space to enumerate is relatively modest. Conversely, cost cuts involves decision variables for all hydropower subsystems and are therefore considerably more challenging to create based on enumeration.

In summary, the creation and management of cuts in ReSDDP is implemented as follows:

- FCF cuts are computed in the backward recursion. These cuts are accumulated, i.e., the number of cuts gradually increases per backward recursion.
- Feasibility cuts are computed before running the SDDP algorithm. The number of feasibility cuts is constant throughout the SDDP run.
- Cost cuts are computed as part of the forward iteration. A user-defined number NC of feasibility cuts are computed per forward scenario and per decision stage.

To allow experimentation with cut management, the ReSDDP model can be run in *continuation mode*. Then, a previously established set of FCF and cost cuts are read as a starting point, and further iterations are run to improve the established strategy. In continuation mode one can choose to turn off further computation of cost cuts.

5.3 Algorithm

The algorithm is outlined in Algorithm 1, where NT is the number of decision periods (weeks), NS the number of sampled inflow scenarios, NC the number of cost cuts to be computed, NW the number of VRES scenarios considered, and NB the number of inflow branches used in the backward recursion. In practical case studies the convergence check is replaced by a visual inspection of the lower bound stabilization and the gap closure between upper and lower bounds after a predefined number of iterations.

Note that NC can be set to an arbitrary number. If both NC and NS is low, the immediate cost function will be underestimated in the early iterations, and one may find that the simulated cost increases along with SDDP iterations. This behavior was discussed in when presenting the numerical examples in Chapter 4, and shown in Fig. 4.6.

while *NotConverged* **do**
- **Forward Simulation**
- **for** $t = 1, .., NT$ **do**
  - **for** $s = 1, .., NS$ **do**
    - **for** $s = 1, .., NC$ **do**
      - Solve **StageProblem** in (5.1) to obtain (e_s^*, r_s^*, c_s^*)
      - **for** $s = 1, .., NW$ **do**
        - Update and solve **CostProblem** in (4.8) to obtain $(\mu^e, \mu^r, \mu^c, \beta^C)$
      - **end**
      - Store cost cut
    - **end**
    - Store state trajectory
  - **end**
- **end**
- *Collect Bounds and Check Convergence*
- **Backward Recursion**
- **for** $t = 1, .., NT$ **do**
  - **for** $s = 1, .., NS$ **do**
    - **for** $b = 1, .., NB$ **do**
      - Solve **StageProblem** in (5.1) to obtain (π^v, π^z, β)
    - **end**
    - Store FCF cut
  - **end**
- **end**

end

Algorithm 1: ReSDDP Algorithm.

Chapter 6

Computational Experiments

This section documents computational experiments that has been performed with the ReSDDP model on a small-scale test system. The intention is to give an overview of the computational performance and the type of results that one can expect from the model. It should not be read as a detailed analysis of the results. In a different study documented in [25], results from ReSDDP applied to the Northern European system for a 2050 scenario are presented.

The first part of the experiments are done without the use of feasibility cuts. In Section 6.6 the experiment is repeated with feasibility cuts to show the impact of more details in the hydropower system. All experiments are run with cost cuts.

6.1 Data and Setup

The test system comprises 4 price areas: 3 with hydropower and one without. An overview of the grid topology and annual production (hydro, wind, solar), other market interactions (price-sensitive demand, thermal generation), and price-insensitive demand is provided in Table 6.1. The test system is set up with 50 weather years representing scenarios for inflow, wind, and solar power. Wind, solar power and comes at hourly resolution, while inflow is at weekly resolution. The data set is available at `https://gitlab.sintef.no/energy/res100/datasets/-/tree/main/4area_res100`.

Area no	Connected to	Hydro	Wind	Solar	Net market	Demand
1	2, 3	2.85	1.64	3.56	-2.32	5.00
2	1, 3	2.26	0.00	0.00	0.00	4.09
3	1, 2	3.36	0.00	0.00	0.05	4.00
4	3	0.00	7.09	0.00	-0.90	4.50
SUM		8.47	8.73	3.56	-3.17	17.58

Table 6.1: Topology, and average annual production (hydro, wind and solar) and demand for the test system. Energy given in TWh.

We ran the ReSDDP model according to the following fixed set up:

- 156 week horizon (NT)
- Weekly decision stages
- 40 sampled inflow scenarios (NS) in the forward simulation
- 5 VRES scenarios (NW)

- 7 points discretizing the residual (NB) in the backward recursion
- Maximum 50 iterations

Three main cases were considered, each with different time discretization within the week: Cases **T8**, **T3**, and **T1**, having 8-, 3-, and 1-hour time resolution within the week, respectively. After computing strategies for each case, a final simulation on 50 fixed inflow and VRES scenarios was performed.

The model was run using the dual simplex algorithm in CPLEX 20.1 on a Intel Core i9-13900K processor with 24 cores, maximum turbo frequency of 5.8 GHz and 32 GB RAM. A total of 21 threads were made available for the program at each run.

6.2 Convergence

In the following we study the convergence properties of the model. Note that a visual approach is taken, leaving deeper analyses of the formal convergence properties for further studies. Fig. 6.1 shows the upper (stapled) and lower boundaries (solid drawn) for all cases. Naturally, the operational cost increases with finer time resolution.

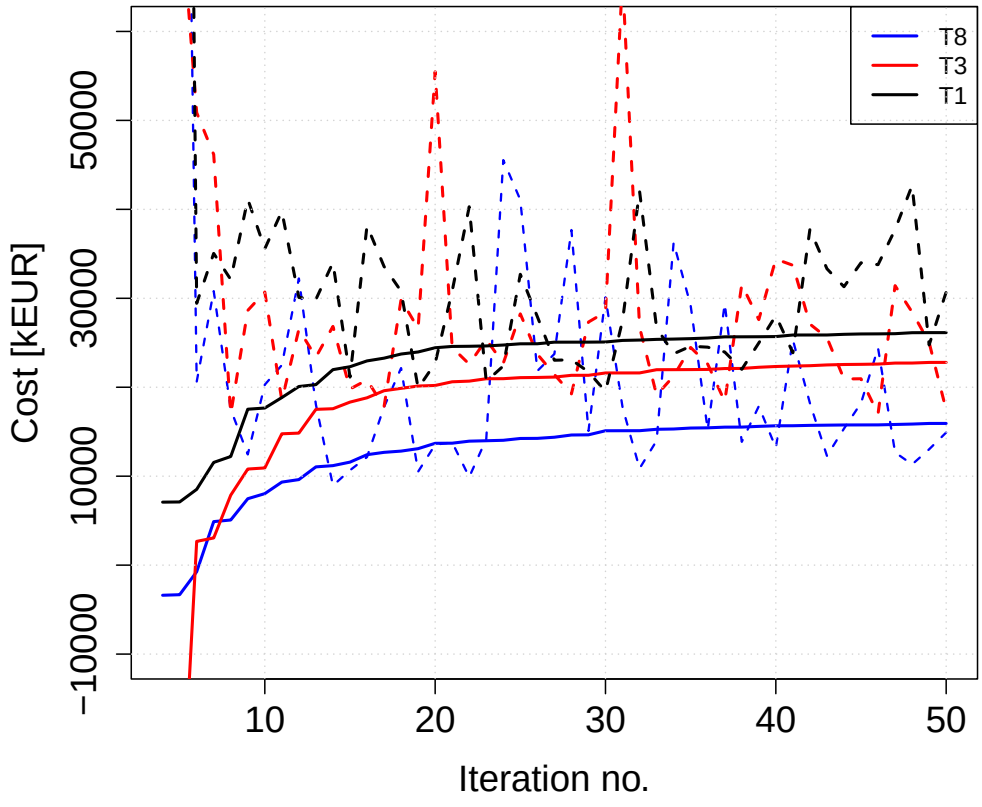


Figure 6.1: Cost gap as a function of SDDP iteration number.

While the lower boundaries seem to gradually stabilize with increasing iteration counter, the upper bounds vary considerably. The variability expresses sampling uncertainty: Using a higher number of forward samples will decrease variability. To study the impact of using a higher number of forward scenarios, a new variant of

case T1 was run, using 400 forward samples. Fig. 6.2 shows the boundaries obtained with 40 (grey) and 400 (black) forward samples. As expected, the variability of the upper bound is considerable lower in the case with 400 scenarios, due to less sampling uncertainty. However, due to the extreme computation time, we do not recommend running ReSDDP model with so many forward scenarios on a desktop computer.

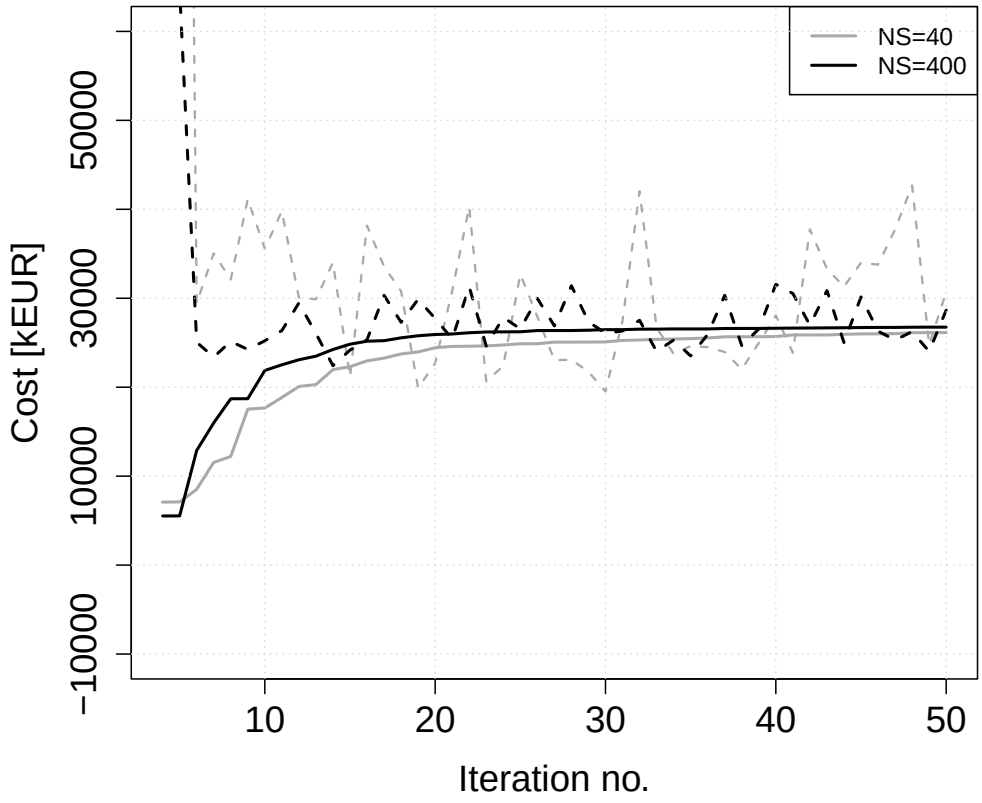


Figure 6.2: Cost gap as a function of SDDP iteration number for case T1 with 40 and 400 forward samples.

While finer parametrization of the SDDP model (more backward openings and forward sampled scenarios) and more iterations may be needed to reach a formal optimum, initial experience from this system indicates that good and relatively stable results are obtained with the current set-up.

6.3 Computation Time

The overall computation time is shown in column four in Tab. 6.2. A first observation is that computations for this test system can be done within reasonable computation times using a modern desktop computer. In the standard SDDP algorithm, majority of the workload is carried out in the backward recursion. In the ReSDDP algorithm it is the other way around. With increasing time resolution, solving cost problems in the forward iterations will dominate the computational time. Recall, that the stage problems solved in the backward recursion only uses, and do not compute, cost cuts.

Fig. 6.3 shows the accumulated time per case spent on forward simulation (stapled) and backward recursion (solid). We emphasize that there is significant potential for improving computational speed by considering: (i) cut management techniques, such as relaxation or filtering of non-redundant FCF cuts; (ii) advanced parallel processing schemes; and (iii) the use of compiled languages.

Case	Forward Average [min]	Backward Average [min]	Total [min]
T1	2.21	0.41	128
T3	0.89	0.36	61
T8	0.67	0.47	56

Table 6.2: Computation times, average per iteration and total per case (in minutes).

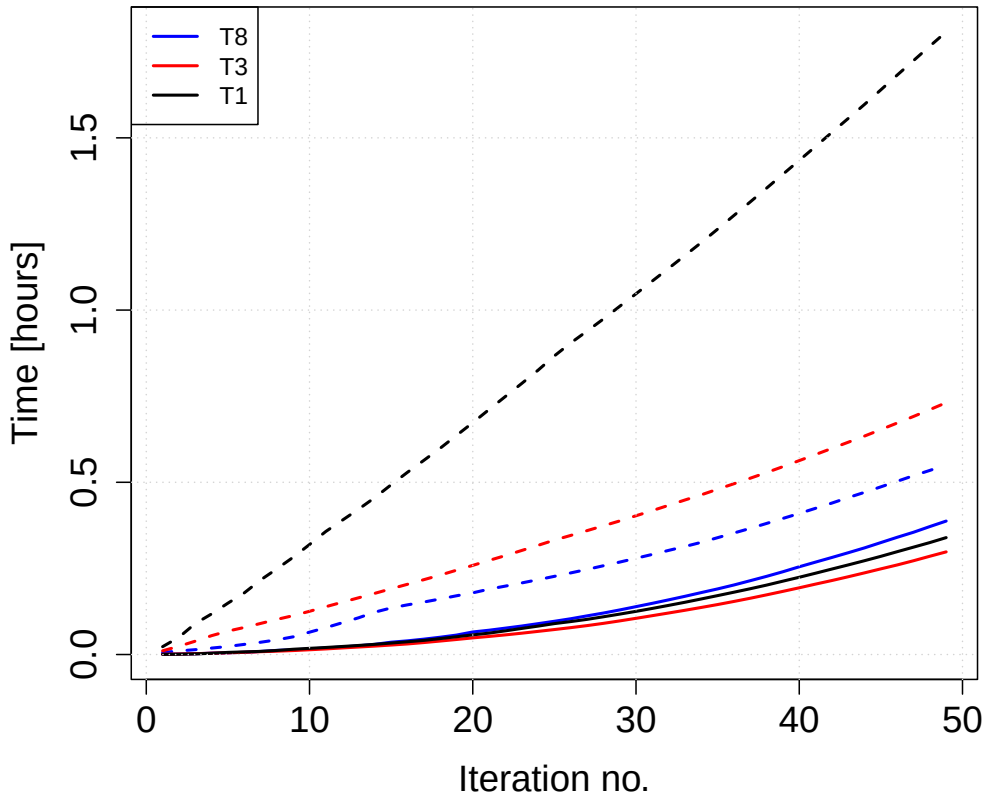


Figure 6.3: Accumulated computation time (in hours) for forward and backward iterations.

6.4 Simulation Results

A total of 50 inflow scenarios were sampled from the stochastic inflow model and combined with the 50 available wind scenarios, ignoring possible correlation between inflow and wind. A final simulation was then conducted along these scenarios using the the computed strategies for cases T1, T3, and T8. The finest time resolution (one hour) was used in all simulations, and we simulate the two first years in the planning horizon. Reservoir trajectories for all simulated scenarios are shown for case T1 in Figs. 6.4. The reservoir trajectories for both of the other cases are similar.

Duration curves for simulated prices for area 3 are shown in Fig. 6.5. The curves are capped at 100 €/MWh to emphasize the lower part. Prices above 100 €/MWh occurs for all cases, typically reaching the curtailment price of 400 €/MWh. The figure indicates that the use of T1 or T3 strategies does not impact the simulated prices much.

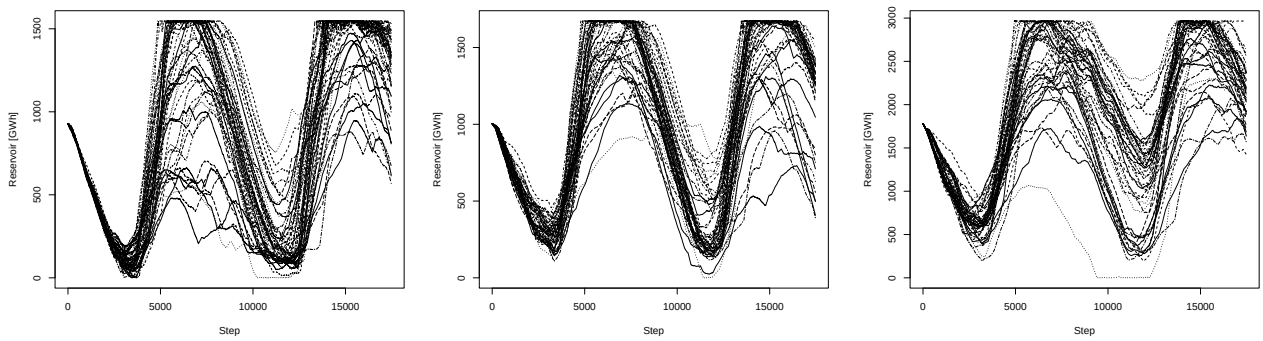


Figure 6.4: Simulated reservoir trajectories for case T1.

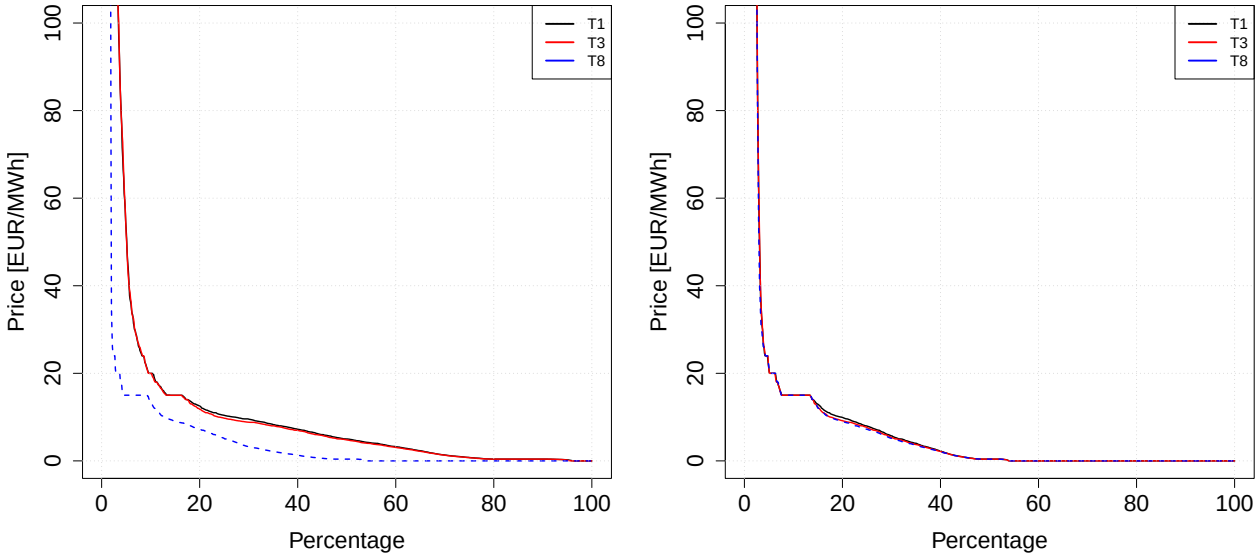


Figure 6.5: Simulated power prices for areas 3 (left) and 4 (right) as duration curves. Prices are capped at 100 €/MWh.

6.5 Water Values

How does fine time resolution impact the strategy (water values)? This question can be addressed by investigating the computed strategies for each of the three cases. Two different techniques can be applied for this purpose, as outlined below.

First, we fix reservoir storages to discrete values and assume average inflow. The FCF cuts in (5.1i) can then be transformed into $\alpha_{t+1} \geq \beta_c^*$, where β_c^* is a constant. Consequently, the cut with highest β_c^* will be binding, and cut coefficients for this cut are interpreted as the water values for the considered combination of fixed reservoir storages. As an assumption, we let the relative reservoir volumes increase proportionally to find the water values for increasing storage volumes.

Fig. 6.6 shows the water values found for all three cases for subsystem 3 for weeks 1, 10, and 20 in the second year of the scheduling period. A first observation is that the values are consistent in the sense that they decrease with increasing storage volume. Although this behavior is expected, it is not guaranteed when using the approach described above. For both weeks 1 and 10 we observe strong incentives to avoid empty-

ing the reservoir, with water values approaching the rationing price (400 €/MWh). The relatively low water values at low reservoir volumes for week 20 can be explained by the lack of cuts computed for this reservoir state. In other words, extremely low reservoir volumes are not considered realistic a few weeks into the snow melting season. Approaching the upper part of the reservoir leads to low water values in all weeks, signaling a pronounced spillage risk. Differences between the three cases seems muted, but remember that axis scales are coarse to capture all values. As a general trend, water values in the most critical zones (i.e. around 50% filling in week 1 and 20% in weeks 10 and 20) are higher with finer time resolution.

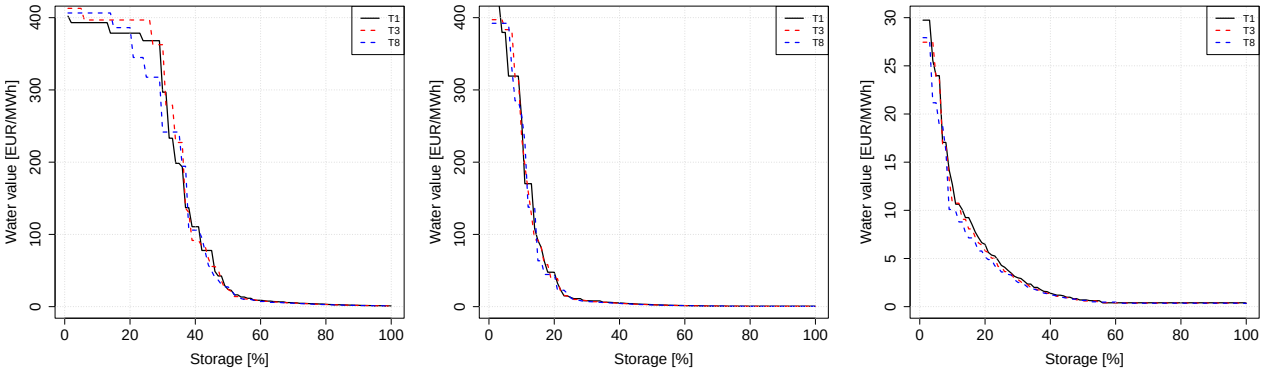


Figure 6.6: Water values for subsystem 3, for weeks 1 (left), 10 (middle), and 20 (right).

Further insight is provided by considering how the water values for a subsystem depends on its own and one of the other subsystem's reservoir volumes. Such 3D plots are provided in Fig. 6.7 for weeks 1 and 20 in the second year of the scheduling period.

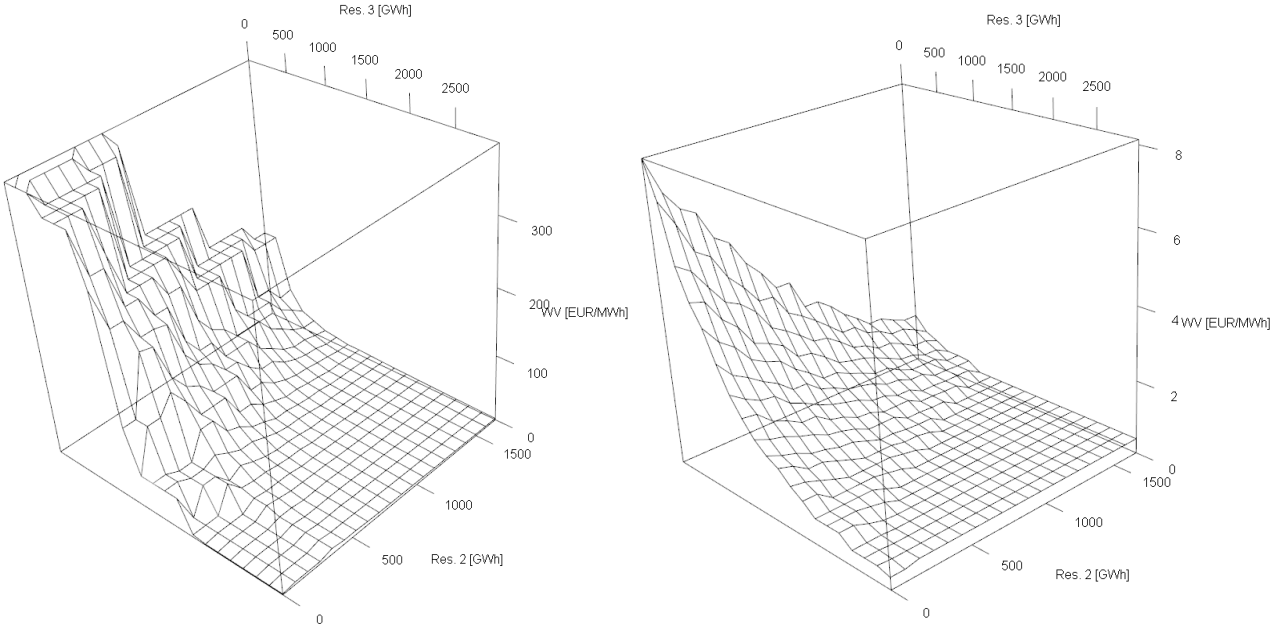


Figure 6.7: Water values for subsystem 3, for weeks 1 (left) and 20 (right).

As a second technique for visualizing water values, we extracted cut coefficients for non-redundant FCF cuts for selected stages. Redundant cuts were removed by applying the elimination procedure outlined in [28] (Section 4.3). The ranges of possible cut coefficients for all subsystems for weeks 1 and 20 in the second year of the scheduling horizon are shown in Figs. 6.8.

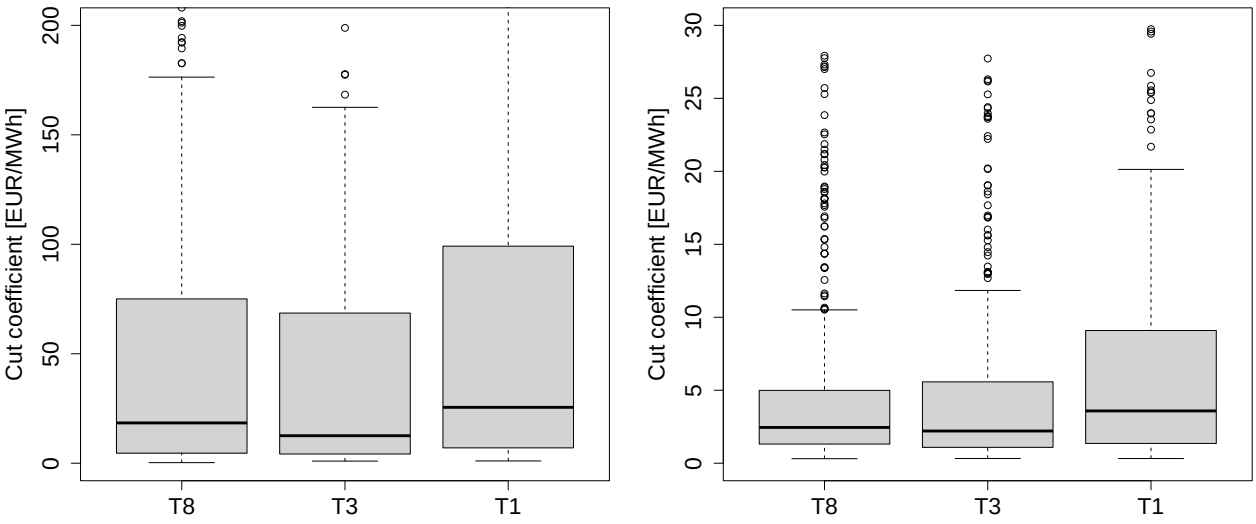


Figure 6.8: Possible water values in weeks 1 (left) and 20 (right). The boxes hold values between the 75th and 25th percentiles, indicating the mean value by a horizontal line.

6.6 Feasibility Cuts

We repeated cases T1 but this time with the use of feasibility cuts. Feasibility cuts were computed prior to the SDDP run and used throughout the iterations. A 3-hour time resolution was used when computing feasibility cuts, and the subsystem decisions related to energy, volume, and ramping capability were discretized in 5 segments. The need for reserve capacity was not considered, and all time-dependent and environmental constraints were deactivated. A total 42, 40, and 35 feasibility cuts were created for each of the three systems. As a consequence, the total computation time increased with approximately 30 % compared to the values reported in Table 6.2.

The strategies were simulated using the same 50 scenarios as for the cases without feasibility cuts. We compare simulation results for case T1 with and without feasibility cuts in Fig. 6.9 and 6.10.

The duration curves for production for the three hydropower subsystems are shown in Fig. 6.9. This figure shows that, although the total energy production per subsystem seems similar, the use of feasibility cuts (black curves) leads to a more constrained operation for subsystems 2 and 3.

Duration curves for simulated power prices for areas 3 and 4 are shown in Fig. 6.10. The use of feasibility cuts further constrains the operation of hydropower, which in turn contributes to a higher price level.

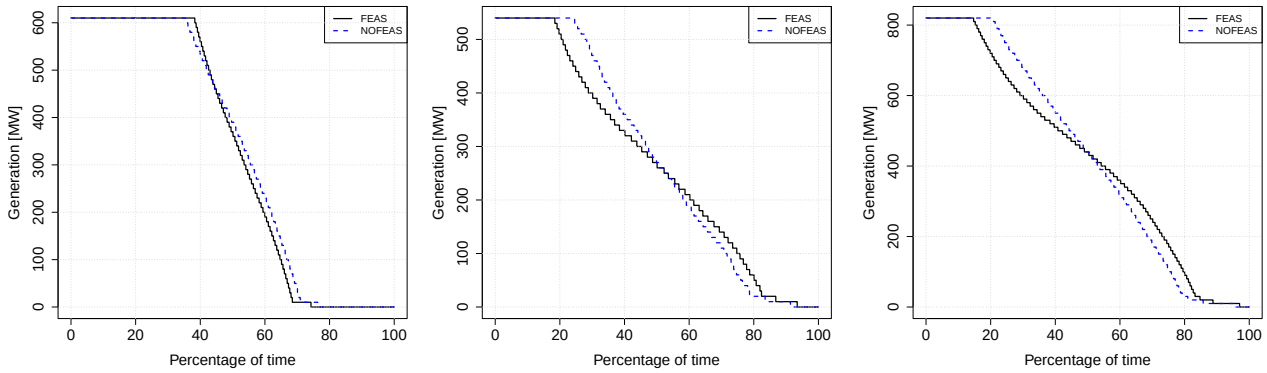


Figure 6.9: Duration curves for production for subsystems 1 (left), 2 (middle), and 3 (right).

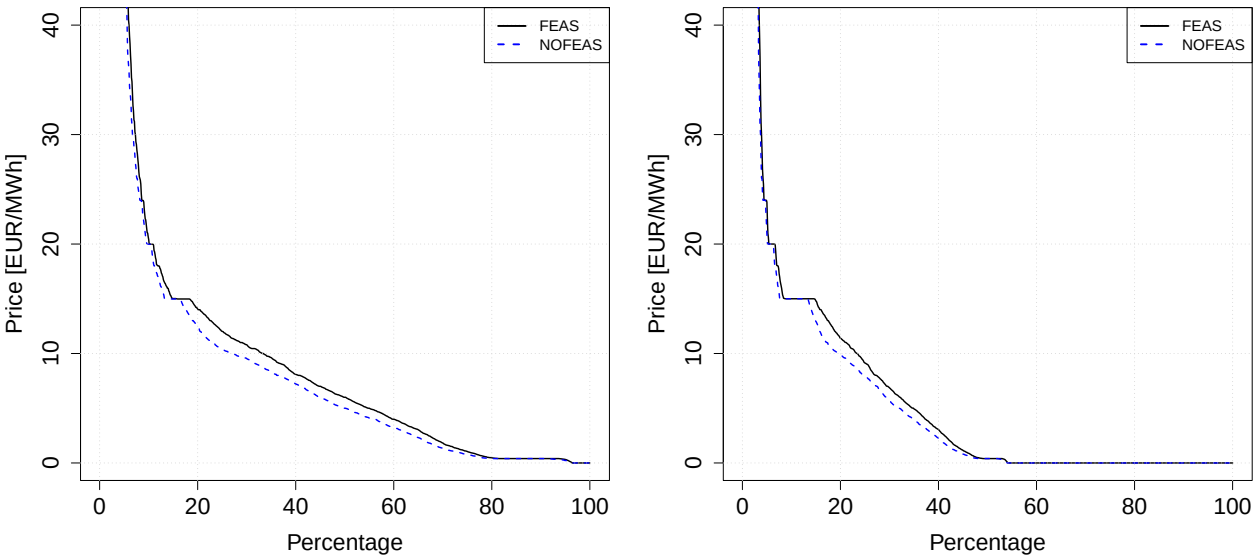


Figure 6.10: Simulated power prices for areas 3 (left) and 4 (right) for case T1 with (black) and without (blue) feasibility cuts. Prices are capped at 40 €/MWh.

Chapter 7

Summary

This report documents the methodological basis for the ReSDDP model. It first introduces the operational planning problem and describes the SDDP algorithm. Subsequently, further decomposition through feasibility cuts and cost cuts are presented, leading to the ReSDDP model algorithm outlined in Chapter5. Finally, the model was applied to a simple test system to demonstrate its computational capabilities and show some of the result types that one can extract from it.

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